GUIDE FOR CONSISTENCY OF THE TITLE AND LABELS

TITLE OF THE WHOLE ACTIVITY: BioQuest: A Journey Toward Research Discovery!

BASECAMP: THE MISSION ORIENTATION

1ST STATION: Moss Quest : Discovering the World of Mosses

2ND STATION: **Mountain Sampling Expedition: Exploring Elevational Distribution of Mosses.**

3RD STATION : Green Grwoth Lab: The In vitro Propapagation of Moss

4TH STATION: Bioactive Secrets: The Phytochemical Screening of Moss

5TH STATION: DEFESE MISSION: THE ANTIBACTERIAL ASSAY OF MOSS

PEAK: DISCOVERY SUMMIT

EVERY STATION NAAY 3 SUBPARTS

1. DISCOVERY STOP – CONTENT PART
2. RESEARCH MISSION – ACTIVITY/ACTIVITIES
3. TRAIL CHECKPOINT – STATION ASSESSMENT/ QUIZ

EVERY AFTER FINISHING THE TRAIL CHECKPOINT NAAY MARECEIVE ANG BATA NA BADGE
STATION 1: RESEARCH BEGINNER BADGE
STATION 2: FIELD SAMPLING PATHFINDER
STATION 3: DATA COLLECTION SPECIALIST
STTAION4: BIOACTIVE CODEBREAKER
STATION 5: DATA DECODING MASTER

SUMMIT: CERTIFICATE WITH THE CITATION: CONGRATULATIONS Emerging Scientific Researcher

SPECIAL REMINDERS

- CONSITENCY SA BACKGROUND LANG SIR EVERY STATION. NOT NECESSARILY NA MAO JUD EMU E BACKGROUND SA NAA DIRI PERU AT LEAST MUMATCH LANG SA GIVEN BACKGROUND SA TOPIC EVERY STATION.
- USE OF PICTURES AND MUSIC AND OTHER RESOURCES WITH NO COPYRIGHT ISSUE
- WITH BACKGROUND MUSIC BUT NOT TOO LOUD SA MGA PARTS NA WALAY GA ESTORYA

1ST PLATE : PINAKACOVER (WITH BACKGROUND MUSIC)



2ND PLATE (AFTER LOGGING IN) STUDENTS WILL CHOOSE THEIR AVATAR
(NOTE: PLEASE INCLUDE GIRLY/FEMININE TYPE OF PANTS AND SHIRTS AND SHOES) REMOVE THE WORD SKIRT IN NUMBER 5



3RD PLATE: AFTER CLICKING START MY ADVENTURE: A MOVING ARROW OR SOMETHING NGA MAGSIGA2 MAGINDICATE NGA E CLICK NILA ANG BASECAMP PARA DIDTO SILA MAGSUGOD



4TH PLATE: ONCE MACLICK ANG BASECAMP MUGAWAS ANG TALKING AVATAR UG MUSTORYA SA WELCOME MESSAGE
(Note: Ang talking avatar lang sa ang mugawas una ayha ng 3 ka clickable ”icons” once muengun na siya na, to better prepare yourselves for this journey READ THE SHORT OVERVIEW, THE GENERAL INSTRUCTIOSN AND THE OBJECTIVES OF THIS MATERIALS.



(SCRIPT FOR THE AVATAR: ENTICING TONE)

Welcome, Young Researcher Explorer!

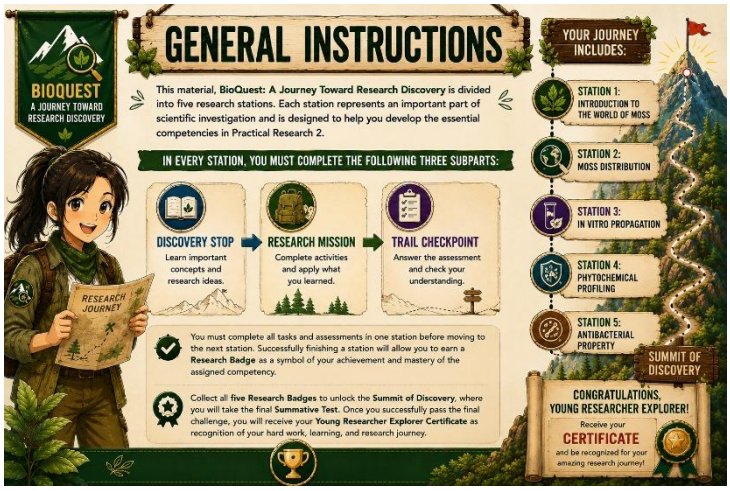
Welcome to **BioQuest: A Journey Toward Research Discovery!** Get ready to begin an exciting adventure filled with exploration, challenges, and scientific discovery. Throughout this interactive learning journey, you will strengthen your research skills, critical thinking, and scientific understanding through fun and engaging activities.

To help you prepare for this journey, read the following: Short Overview, General Instructions, and Instructional Objectives.

5th PLATE: E CLICK SA STUDENTS AND SHORT OVERVIEW (DAPAT NAAY CLIKABLE BUTTON NA MAKABALIK SILA SA 4TH PLATE.) DIRI IF POSSIBLE NAAY VOICEOVER SA SHORT OVERVIEW PERU PWEDE SIYA E OFF IF NAA STUDENTS DILI GANHAN MAMINAW



6TH PLATE : THE GENERAL INSTRUCTION (DAPAT NAAY CLIKABLE BUTTON NA MAKABALIK SILA SA 4TH PLATE. .) DIRI IF POSSIBLE NAAY VOICEOVER SA INSTRUCTIONS PERU PWEDE SIYA E OFF IF NAA STUDENTS DILI GANHAN MAMINAW



7TH PLATE: THE INSTRUCTIONAL OBJECTIVES (NAAY CLICKABLE ICON NA MABALIK SA 4TH PLATE) .) DIRI IF POSSIBLE NAAY VOICEOVER SA INSTRUCTIONAL OBJECTIVES PERU PWEDE SIYA E OFF IF NAA STUDENTS DILI GANHAN MAMINAW



8TH PLATE: DIRI MAGSTORYA ANG AVATAR



(SCRIPT FOR THE AVATAR: EXCITING TONE)

Your mission briefing is now complete!
Get ready to explore, investigate, and discover as you begin your exciting journey toward the **Summit of Discovery**. Adventure, challenges, and scientific discoveries await you, Young Researcher Explorer!

9TH PLATE: MAGSUGOD NA SILA DIRI EVERY STATION (PRETEST FIRST)



10TH PLATE: automatic makabalo sila Sa ila score after taking all the items, but answers should not be revealed



QUESTIONS:

1. **What laboratory process is used to identify bioactive compounds present in mosses?**

A. Soil Profiling
B. Ecological Mapping
C. Photosynthesis Analysis
D. Phytochemical Screening

2. Which of the following is an example of a sample?

A. All mosses on the mountain
B. All organisms in the ecosystem
C. Every plant species in the forest
D. Mosses collected from one small area

3. A student needs to measure the shoot height of a moss sample accurately. Which instrument should be used?

A. Ruler B. Camera C. Beaker D. Stopwatch

---

4. Which statistical tool is commonly used to determine the average value?
   A. mode B. mean C. median D. ANOVA

5. Which step should researchers accomplish first before organizing gathered data?
   A. Identifying the data that must be gathered
      a. Recording all experimental observations carefully
      C. Computing the final interpretation of the results
      D. Organizing the tables and graphical presentations

6. **Which research design involves manipulating variables under controlled laboratory conditions?**
   A. Descriptive Research
   B. Correlational Research

C. Experimental Research
D. Quasi-Experimental Research

7. In the antibacterial assay investigation, what will be measured by the researchers?
   A. The amount of moss collected
   B. The temperature inside the room
   C. The diameter of the zone of inhibition
   D. The number of bacteria found in the plate

---

8. What does the mean represent in scientific investigations and experiments?
   A. The middle value after arranging all scores
   B. The conclusion written after data interpretation
   C. The smallest value among all measurements taken
   D. The average obtained from repeated measurements

---

9. Which study title is an example of descriptive research?
   A. Identifying moss species found on tree trunks
   B. Comparing moss abundance in two forest areas
   C. Testing the effect of nutrient solutions on moss growth
   D. Determining the relationship between moisture and moss diversity

10. A group collected measurements using different instruments incorrectly. What is the most likely effect on their investigation?
    A. Faster moss growth
    B. More colorful cultures
    C. Less reliable research data
    D. More laboratory equipment

11. A researcher tests the antibacterial activity of moss extracts against bacteria. Which research design should be used?
   A. Experimental Research
   B. Descriptive Research
   C. Correlational Research
   D. Quasi-Experimental Research

12. Which example below is considered quantitative data in the investigation?
    A. Name of the researcher conducting the study
    B. Type of bacteria used during experimentation
    C. Color appearance of the prepared agar plates
    D. Measurement of the inhibition zone in millimeters
13. Why is choosing the correct instrument important in scientific investigations?
    A. To finish slowly
    B. To avoid observations
    C. To make the laboratory colorful
    D. To collect accurate scientific data
14. Why do researchers usually perform several replicates in an experiment?
    A. To shorten the duration of the investigation
    b. To increase the reliability of gathered results
    C. To reduce the preparation of laboratory tools
    D. To replace the original hypothesis completely

15. Why is organizing collected data in tables important during scientific investigations?

A. To make the report colorful
B. To collect the data in a correct way
C. To make the report longer and better
D. To compare and analyze data easily

16. What is the main purpose of planning data analysis in a research investigation?
A. To organize and interpret data scientifically
B. To shorten and simplify the whole experiment
C. To decorate and improve the research paper
D. To increase the number of laboratory materials

For numbers 17–20. Refer to the hypothetical data below about the *in vitro* propagation of two moss species.

Researchers observed the number of new shoots produced by two moss species during 4 weeks of *in vitro* propagation.

Week	Moss Species A	Moss Species B
Week 1	5 shoots	7 shoots
Week 2	8 shoots	11 shoots
Week 3	11 shoots	15 shoots
Week 4	14 shoots	20 shoots

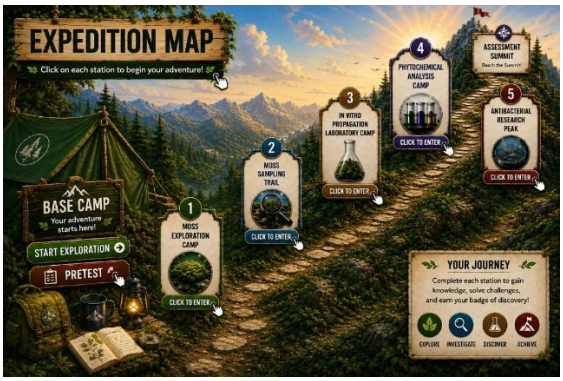
17. Which statement best explains the trend shown by Moss Species B from Week 1 to Week 4?
A. Its growth changed randomly with no clear pattern
B. Its growth steadily increased over the 4-week period.
C. Its number of shoots gradually decreased each week.
D. . Its growth remained constant throughout the experiment.

18. Which type of graph is most appropriate for presenting the growth of the two moss species over 4 weeks?
A. Pie graph
B. Bar graph
C. Line graph
D. Scatter plot

19. Which statement best compares the growth of Moss Species A and Moss Species B during the 4-week *in vitro* propagation experiment?
A. Both moss species produced the same number of shoots each week.
B. Moss Species B consistently produced more shoots than Moss Species A.
C. Moss Species A consistently produced more shoots than Moss Species B.
D. Moss Species A and B showed decreasing growth throughout the experiment.

20. If the experiment continued for two more weeks and the trend remained the same, what is the most reasonable prediction for Moss Species A?
A. Its number of shoots would suddenly decrease
B. Its growth would likely continue increasing gradually.
C. Its growth would remain exactly at 14 shoots forever.
D. Its growth would become lower than Week 1 results.

PLATE 11: CLICKABLE ANG STATION 1



12TH PLATE : INTRODUCTORY PART OF EACH STATION. CHANGE TITLE TO **Moss Quest : Discovering the World of Mosses (Isulod ning welcome....Gagmayan lang ang clickable subparts para maegu ang welcome part), Naay avatar magestroya sa welcome young explorers.....**



Welcome, young explorers! In this first station, you will begin your mountain research journey by discovering the fascinating world of mosses.
As you investigate their characteristics, habitats, and ecological importance, you will also begin thinking like real researchers by learning how to:

- Choose an appropriate quantitative research design

Through this exploration, you will discover that every successful scientific investigation starts with a well-planned research design that guides the entire study.

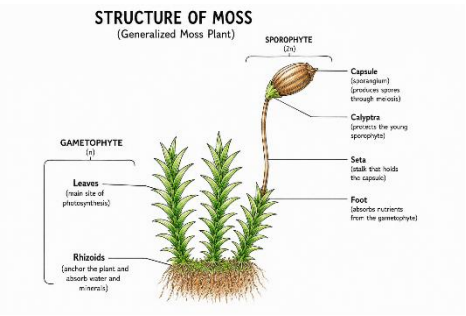
12TH PLATE: STATION 1 DISCOVERY STOP CONTENT (place the content at the blank space) if possible unta sir kanang mag scroll lang within the same page ang content kay medyu taas2 siya. Tas if Makaya pabutangan nako ug voice over sir.Peru ang voiceover pwede sad siya mapalong in case nga nay studyante dili ganahan maminaw ug tingug kay mas gusto niya magbasa ra.



STATION1 DISCOVERY STOP (CONTENT)

Mosses are small, non-vascular plants that belong to a group of plants called **Bryophytes.** This group of plants are unique as they do not have true roots, stem or vascular tissues such as xylem and phloem. Instead, they possess root-like structure called rhizoids that help them attach to the surfaces and absorb moisture from its surroundings. Due to their unique structure, they commonly grow in moist and shaded environments such as the forest floor, rocks, trees, tree trunks and riverbanks as they required water for reproduction and survival.

Despite their small size, mosses contribute significantly to environmental stability and biodiversity in an ecosystem. They help retain moisture in the soil, reduce erosion, recycle nutrients, and provide microhabitats for microorganisms and small invertebrates. Mosses are also considered excellent bioindicators, which means that they can be used to monitor environmental conditions such as pollution and climate change as they are highly sensitive to the changes in their surroundings (Sabovljević et al., 2016). Thus, their ecological importance makes them valuable organisms in environmental and biodiversity studies.



Aside from their ecological functions, mosses are also gaining attention because of their great potential in the field of medicine. Recent studies have shown that some moss species have various secondary metabolites such as alkaloids, terpenoids, phenolic compounds, and steroids to name a few. More interestingly, these compounds are found to have antibacterial, antifungal, antioxidant, and anti-inflammatory activities (Asakawa et al., 2018).

Because of the unique and interesting characteristics of mosses, many scientific studies can be conducted in this field. For example, researchers study mosses to understand their distribution, diversity, ecological importance, habitat specificity, and possible medicinal properties to name a few. Since these studies have different goals and methods of investigation, it is important for researchers to choose the most appropriate quantitative research design.

A **Quantitative Research Design** is a systematic way of collecting and analyzing numerical data to produce reliable scientific results. There are four main types of quantitative research as shown in the table below:

There are 4 Main Types of Quantitative Research

Type of Quantitative Research	**Description**	**Key Question it answers**	**When to use**	**Sample Studies**
Descriptive Research	Describes existing conditions, characteristics, or situations through observation and data collection.	What is present or happening?	Use when the study only aims to describe or identify something.	• Counting the number of mosses growing on rocks • Assessing the distribution of moss species across different elevations of a mountain ecosystem
Correlational Research	Determines whether a relationship exists between two variables without manipulating them.	Are these two variables related?	Use when the study aims to determine whether two variables are related but no groups or treatments are compared	• Studying if colder areas have more mosses • Determining the relationship between elevation and moss species distribution in a mountain ecosystem
Quasi-Experimental Research	Compares two or more existing groups or conditions to observe possible effects without	Which condition or group show better results?	Use when comparing groups or environments where not all variables can be controlled.	• Comparing moss growth on tree bark and rocks • Comparing moss abundance and diversity between disturbed and undisturbed forest areas

	complete control or random grouping.			
Experimental Research	Tests cause-and-effect relationships by manipulating variables under controlled conditions.	What happens if a treatment is applied?	Use when treatments or variables are applied in a controlled setup.	• Testing moss growth in different nutrient solutions • Evaluating the effectiveness of moss extracts at different concentrations against bacterial growth

Again, it is important to note that different scientific investigations require different research designs depending on the purpose of your study. Understanding these quantitative research designs will help you conduct organized, accurate, and reliable scientific investigations.

AFTER CONTENT CLICKABLE ANG RESEARCH MISSION



13TH PLATE: STATION 1 RESEARCH MISSION. (Sa diri na activity if makaya naay murag heart nga mugawas kada makacheck ang bata sa 10 items tas if mahuman sila naa dayun mugawas na moss depende sa ilang score: 0-4 kamatyunon na moss, 5-7 healthy peru nipis ug mubo na moss, 8-9 health tagas ug baga na moss, 10 naay capsule na moss) . NAAY BUTTON MAKASTART SA ACTIVITY



- Mission Story

A rare moss species inside the virtual bryophyte laboratory is slowly losing its health. To restore the moss and help it produce a healthy capsule, you must correctly identify the appropriate quantitative research design in each challenge.

Every correct answer restores the moss health meter. If you get a perfect score, the moss will fully grow and produce a mature capsule!

14TH PLATE: ACTIVITY PROPER



Directions:
Tap the colored box that represents the correct quantitative research design.

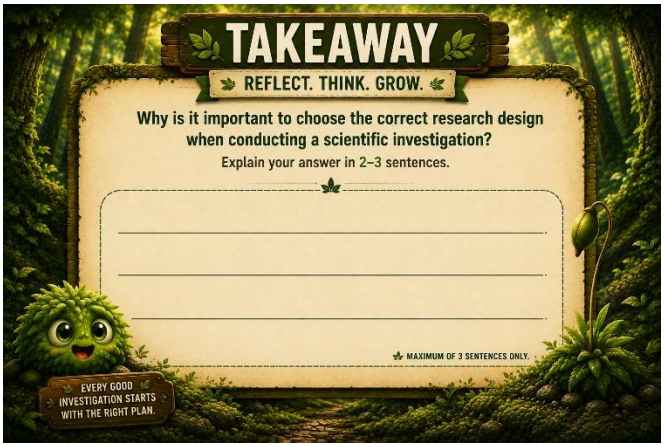
🟦 Descriptive Research Design
A 🟩 Correlational Research Design
🟧 Quasi-Experimental Research Design
🟥 Experimental Research Design

1. A researcher identifies the different moss species growing on rocks near a river.
🟦 🟩 🟧 🟥
2. A researcher studies whether colder places have greater moss abundance.
🟦 🟩 🟧 🟥
3. A researcher compares moss growth in shaded and sunny areas.
🟦 🟩 🟧 🟥
4. A researcher tests the effect of different nutrient solutions on moss growth inside the laboratory.
🟦 🟩 🟧 🟥
5. A researcher determines the relationship between elevation and moss species distribution in a mountain ecosystem.
🟦 🟩 🟧 🟥
6. A researcher compares moss abundance between disturbed and undisturbed forest areas.
🟦 🟩 🟧 🟥
7. A researcher evaluates the antibacterial activity of moss extracts against bacteria.
🟦 🟩 🟧 🟥
8. A researcher counts the number of moss colonies growing on tree trunks.
🟦 🟩 🟧 🟥
9. Determining the relationship between elevation and moss species distribution in a mountain ecosystem
🟦 🟩 🟧 🟥
10. Testing the antibacterial activity of moss extracts against bacterial growth under laboratory conditions
🟦 🟩 🟧 🟥

☑ **Answer Key**

1. Descriptive Research
2. Correlational Research
3. Quasi-Experimental Research
4. Experimental Research
5. Correlational Research
6. Quasi-Experimental Research
7. Experimental Research
8. Descriptive Research
9. Correlational Research
10. Experimental Research

15th PLATE: TAKEAWAY QUESTION. STUDENTS WILL WRITE THEIR ANSWER IN A BOX.



Correct Answer:

Choosing the correct research design is important because it helps researchers gather accurate and reliable data. Different studies have different objectives, so the appropriate research design helps organize the investigation properly.

Score	Condition
5 Points	Answer clearly explains the importance of choosing the correct research design and includes key ideas such as accurate data, reliable results, organization, or proper investigation. Response is clear.
4 Points	Answer explains the importance but lacks one important idea or has minor clarity issues.
3 Points	Answer shows partial understanding but explanation is incomplete or too general.
2 Points	Answer is very short, unclear, or only repeats words from the question.
1 Point	Answer shows very little understanding of the topic.

16TH PLATE: CLICKABLE NA ANG TRAIL CHECKPOINT FOR THE ASSESSMENT PART



17TH PLATE: TRAIL CHECKPOINT. (PWEDE RA ISA, 2 OR 3 KA QUESTION MAIGU SA USA KA SLIDE PARA DAGKO RA ANG FONT . PWEDE MASCORLL DOWN PARA MA VIEW ANG UBAN QUESTION OR NEXT SLIDES. IKAW RA BAHALA SIR. CLICKABLE ANG LETTERS UG AUTOMATIC MAKABALO SILA SA SCORE



1. Which quantitative research design is used when a study only describes existing conditions or characteristics?
A. Descriptive Research
B. Experimental Research
C. Correlational Research
D. Quasi-Experimental Research

---

2. A researcher studies whether colder places have greater moss abundance. What research design is most appropriate?
A. Descriptive Research
B. Correlational Research
C. Experimental Research
D. Quasi-Experimental Research

---

3. Which research design is best used when comparing moss growth in shaded and sunny areas?
A. Experimental Research
B. Descriptive Research
C. Correlational Research
D. Quasi-Experimental Research

---

4. Which research design involves manipulating variables under controlled laboratory conditions?
A. Descriptive Research
B. Correlational Research
C. Experimental Research
D. Quasi-Experimental Research

---

5. What makes correlational research different from descriptive research?
A. It allows to compare groups or environments.
B. It studies relationships between variables.
C. It uses treatments and experiments.
D. It only describes existing conditions.

---

6. A researcher tests the antibacterial activity of moss extracts against bacteria. Which research design should be used?
A. Experimental Research
B. Descriptive Research
C. Correlational Research
D. Quasi-Experimental Research

---

7. Which statement best describes quasi-experimental research?
A. It describes existing conditions only.
B. It determines relationships between variables.
C. It compares groups or conditions without complete control of variables.
D. It always uses laboratory experiments and equipment which involves manipulation of variables.

---

8. Which pair of variables is MOST likely studied in correlational research?
A. Moss species and moss color
B. Sunny area and shaded area
C. Elevation and moss abundance
D. Nutrient solution and laboratory setup

---

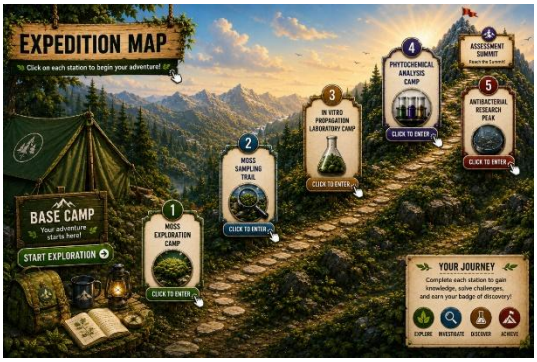
9. Which study title is an example of descriptive research?
A. Identifying moss species found on tree trunks
B. Comparing moss abundance in two forest areas
C. Testing the effect of nutrient solutions on moss growth
D. Determining the relationship between moisture and moss diversity

---

10. Why is it important to choose the correct research design in a scientific study?
A. To make the study longer and better
B. To avoid long duration of collecting data

C. To produce organized and reliable results
D. To reduce the number of variables and conditions

18th plate: Balik diri sa expidition map peru dapat naay progress chart nga nahuman na sila sa first station. Clickable na ang second station



19th PLATE: MAOPEN AND SECOND STATION. TITLE **Mountain Sampling Expedition: Exploring Elevational Distribution of Mosses. (Gagmayan lang ang clickable subparts para masulod ning naa sa arrow), NAAY AVATAR MAGSTORYA SA GREETINGS, MOUNTAIN RESEAERCHERS....**



Greetings, mountain researchers! In this station, you will explroe across different elevations of the mountain to investigate how mosses are distributed from lowland to highland environments.
As you explore moss communities along the mountain trail, you will also learn how researchers carefully gather samples in the field by learning how to:

- Describe appropriate sampling procedures
- Identify the population and sample in a research study

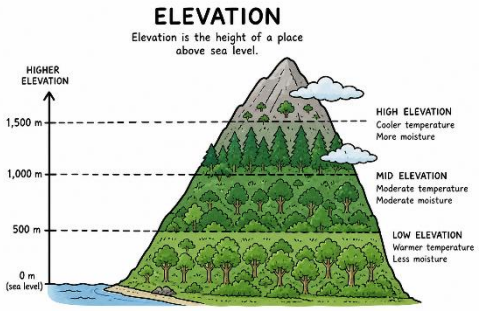
Throughout this expedition, you will realize that proper sampling is essential in obtaining accurate and reliable research data from natural environments.

20TH PLATE: MAOPEN AND DISCOVERY STOP. SULOD NATO DIRI ANG CONTENT SA STATION 2. (Content details below. Dapat naay mo voice over sa content peru pwede gihapon siya ma on and off if ever naay dili gusto ug naay tingug.



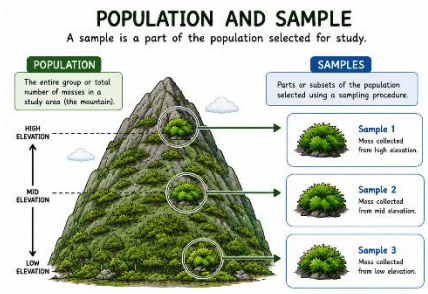
STATION 2 CONTENT: MOSS DISTRIBUTION

Mosses commonly grow in moist and shaded environments and are found to be highly sensitive to environmental conditions such as temperature, humidity, sunlight, and water availability. Because of these characteristics, mosses are often used by scientists as indicators of environmental change. One important factor that affects the growth and distribution of mosses is elevation.



Elevation refers to the height of an area above sea level usually expressed in masl (meters above sea level). As elevation increases, environmental conditions also change. Higher elevations usually experience cooler temperatures, stronger winds, and greater moisture availability compared to lower elevations. These environmental differences influence the types of mosses that can survive and grow in a particular habitat. Some moss species are commonly found in lowland forests, while others thrive only in higher mountainous areas.

Scientists examine how moss species vary from one elevation zone to another to determine how environmental factors affect their growth and survival. However, studying all mosses found in an entire mountain ecosystem is difficult and time-consuming. Because of this, researchers use scientific methods in selecting representative groups of organisms for observation and study. This process involves the use of samples abd sampling procedures.

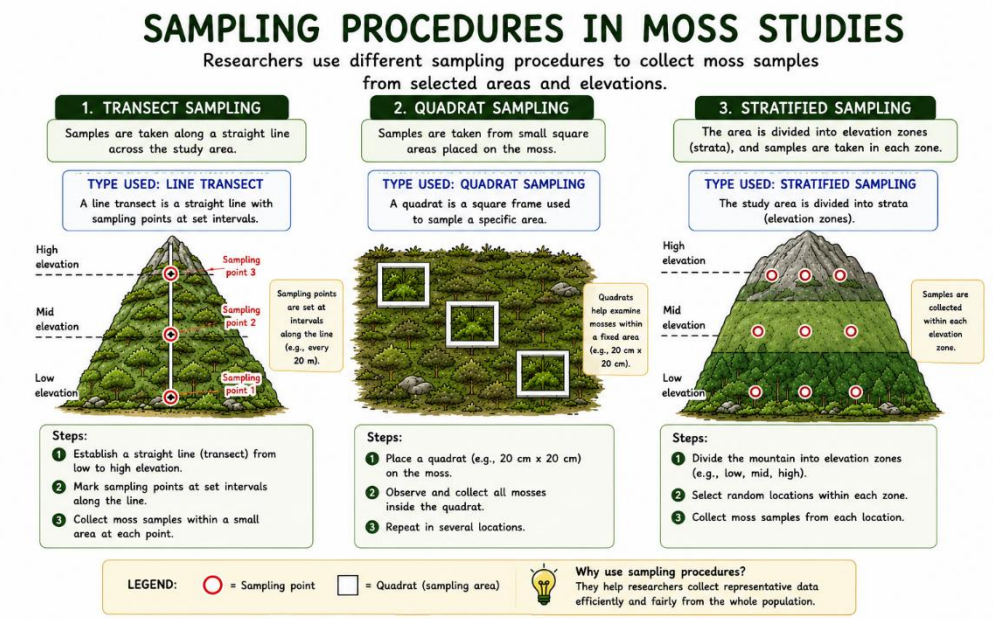


A **sample** refers to a portion or representative group taken from a larger population for analysis. In the study of moss distribution, the entire moss community found in the mountain serves as the population, while the moss colonies gathered from selected sampling areas serve as the samples. These samples provide information about the characteristics of mosses present in different elevations.

To obtain reliable samples, researchers use a sampling procedure. A **sampling procedure** refers to the systematic method used in selecting and collecting samples from a study area. In moss distribution studies, researchers carefully identify specific sampling sites from low, mid, and high elevations to ensure that the collected data properly represent the entire population. Different sampling techniques such as transect sampling, quadrat sampling, and stratified sampling may be used to organize data collection. Through proper sampling procedures, researchers can gather accurate and reliable data that help explain the elevational distribution of mosses.

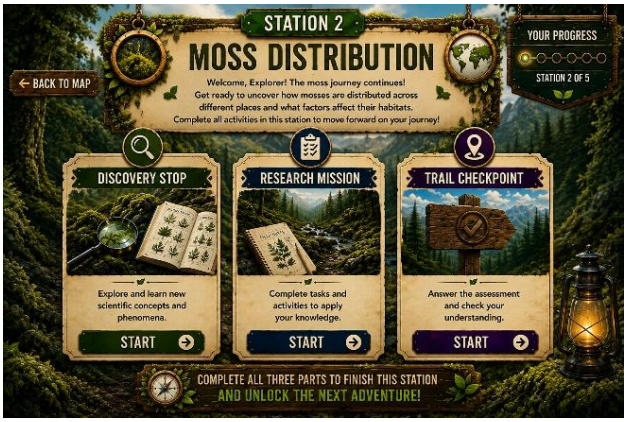
Researchers use different sampling procedures in studying moss distribution. One commonly used method is **transect sampling**. In this method, researchers establish a straight line across an area and observe mosses found along the path. This allows scientists to identify changes in moss distribution from lower to higher elevations. Another method is **quadrat sampling**, where a square frame called a quadrat is placed on selected areas to measure moss abundance, coverage, and diversity within a specific space. **Stratified sampling** may

also be used by dividing the study area into elevation categories to ensure that all elevation zones are represented during data collection.



In actual ecological and biodiversity studies, researchers may combine two or more sampling procedures to gather more accurate and reliable data. Combining sampling methods helps ensure that the collected samples properly represent the study area. One common example is the combination of transect sampling and quadrat sampling. In this method, researchers first establish a transect line across different elevations and then place quadrats at specific intervals along the line to collect moss samples. The choice of sampling procedure depends on the research problem, study area, environmental conditions, and available resources. Through the proper selection and combination of sampling procedures, researchers can obtain more organized and representative data in studying the elevational distribution of mosses.

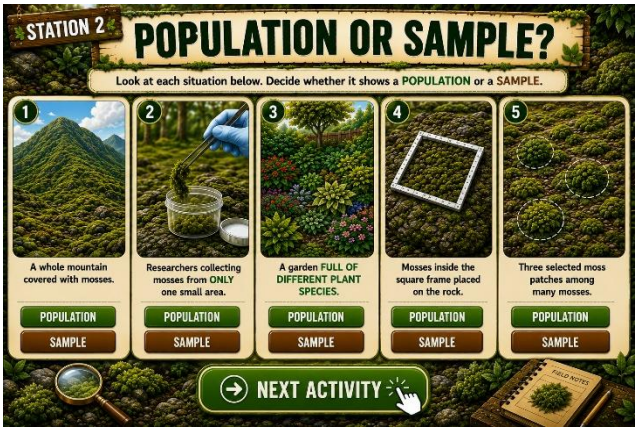
21 PLATE: MAKE RESEARCH MISSION CLICKABLE



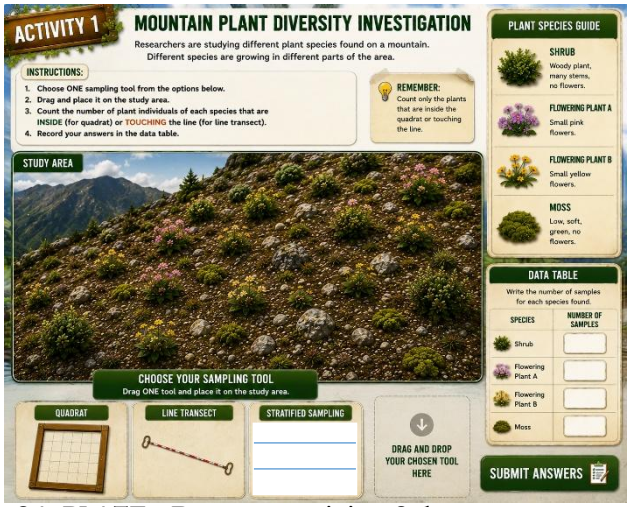
MUGAWAS ANG RESEARCH MISSION INTRODUCTION. NAAY BUTTONN PARA MAKATSART SA ACTIVITY



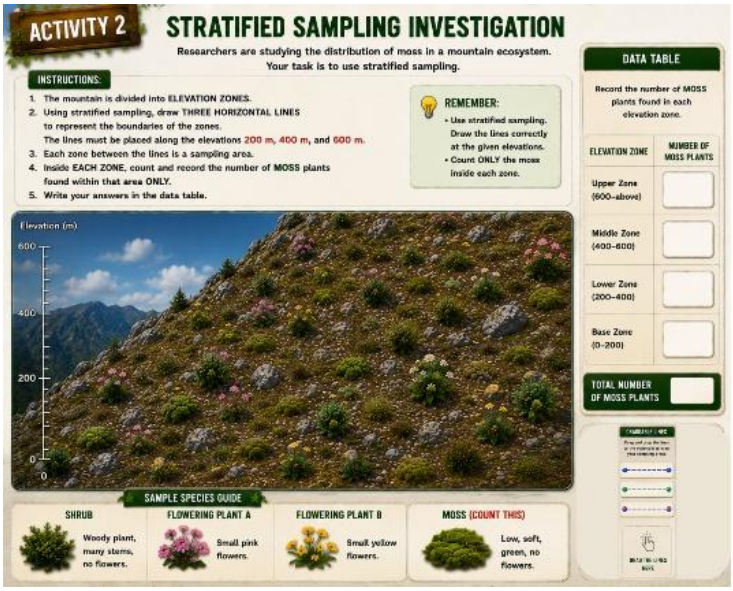
22 PLATE: MUGAWAS ANG MAIN ACTIVITY FOR STATION (The Main activity is divided into 2 parts) (Automatic muidentify if correct or wrong ila answer)



23 PLATE: (If possible sir, clickable ang mountain para whole mountain view nag makita. Tas automatic muhatag ug correct answer sa number of samples



24 PLATE: Remove activity 2 kay sumpay rani sa previous activity (unta maclickable ang mountain para mudako ang view sa mountain)



25TH PLATE: REMOVE LANG ACTIVITY 3 and reflect and answer na word kay sumpay rani siya sa last activity. ADD A CLICKABLE BUTTON FOR A TAKEAWAY QUESTION.

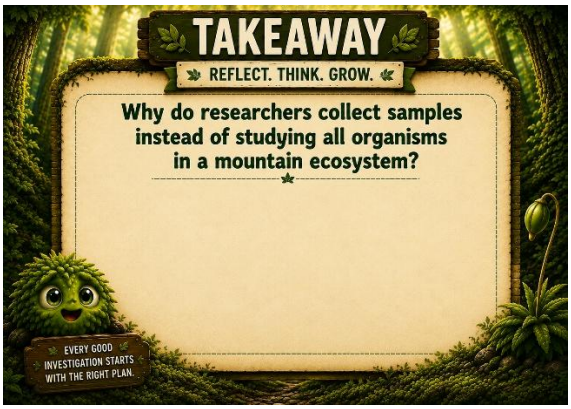


Ang question 1 is : What type of sampling technique is used on the previous activity?
Answer: Stratified Sampling

Answer Question 2: Because the mountain was divided into different elevation/sections areas using horizontal lines. The moss samples were counted separately in each sampling area or zone.

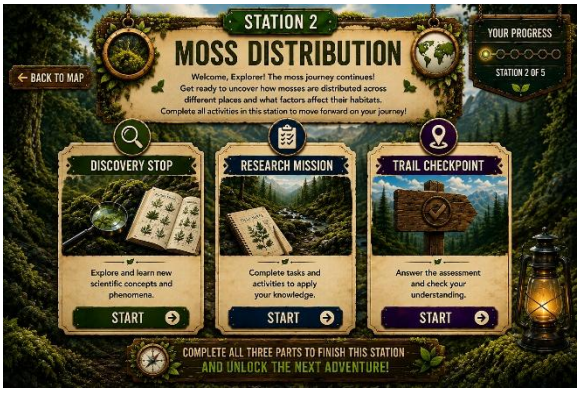
Note: for this, give automatic score using the following conditions:
A score of **5 points** is given if the learner clearly explains that the mountain was divided into different elevation zones or areas and that mosses were counted separately in each zone and no wrong spelling of words are used. A score of **4 points** is given if the learner mentions dividing the area and counting samples but gives a less detailed explanation. A score of **3 points** is given if the learner shows partial understanding by mentioning different areas or groups only. A score of **2 points** is given if the learner only mentions sampling or counting without explaining zones or stratification. A score of **1 point** is given for a very short, unclear, or weak answer. A score of **0 point** is given if the answer is blank, irrelevant, or completely incorrect.

26TH PLATE: (NAAY CLICKABLE ITEM PARA MAKABALIK SA NEXT PLATE)

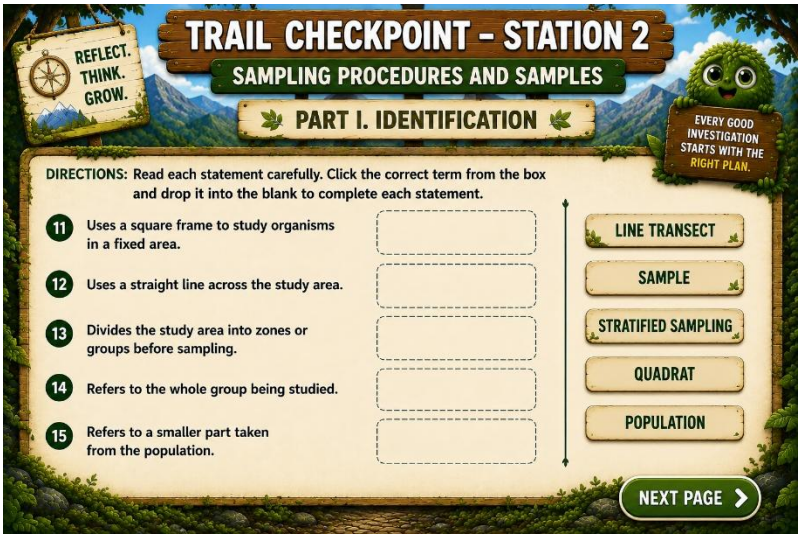


Researchers collect samples because studying all organisms in a mountain ecosystem is difficult, time-consuming, and requires many resources. Samples help researchers gather information that represents the whole population in a faster and more organized way.

27TH PLATE: CLIACKABLE NA ANG TRAIL CHECKPOINT READY SA ASSESSEMNT PART



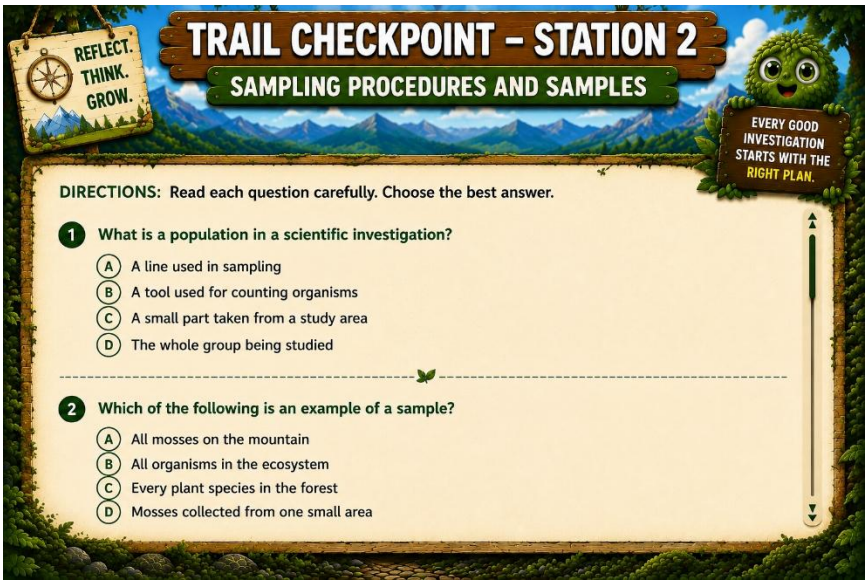
28TH PLATE: MUGAWAS ANG ASSESSMENT ((turn number 11 to 1, 12 to 2..etc.)



Correct Answers:

1. Quadrat
2. Line Transect
3. Stratified Sampling
4. Population
5. Sample

29th PLATE: TRAIL CHECKPOINT STTAION 2 PART 2(MULTIPLE CHOICE). IKAW NA BAHALA SIR UNSAON NA INTERACTIVE ANG MULTIPLE CHOICE



Choose the correct answer.

1. What is a population in a scientific investigation?
A. A line used in sampling
B. The whole group being studied
C. A tool used for counting organisms
D. A small part taken from a study area

---

2. Which of the following is an example of a sample?
A. All mosses on the mountain
B. All organisms in the ecosystem
C. Every plant species in the forest
D. Mosses collected from one small area

---

3. Which sampling tool uses a square frame?
A. Quadrat
B. Line Transect
C. Belt transect
D. Stratified sampling

---

4. In a line transect, researchers count organisms that:
A. touch the line
B. are outside the study area
C. are found only on rocks
D. are inside a square frame

---

5. Why do researchers use sampling techniques?
A. To avoid collecting data
B. To study only one organism
C. To make investigations longer
D. To gather organized and reliable data

6. What type of sampling divides the study area into zones or layers?
A. Random sampling
B. Quadrat sampling
C. Stratified sampling
D. Line transect sampling

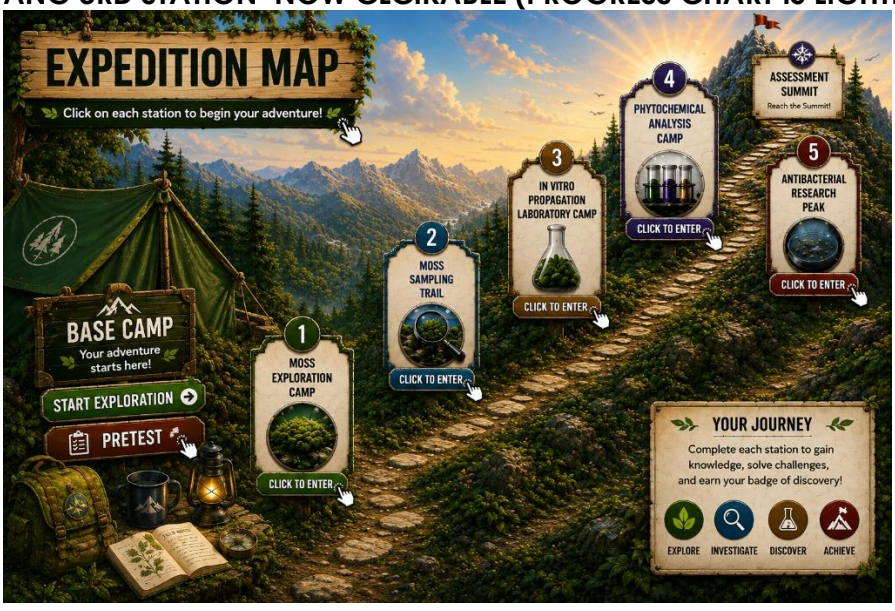
7. In the activity, the mountain was divided using:
A. circles
B. vertical lines
C. horizontal lines
D. square quadrats

8. In moss distribution studies, what serves as the population?
A. Moss colonies gathered from quadrats
B. Mosses collected from selected elevations
C. Entire moss community within the mountain
D. Researchers studying environmental conditions

9. What is the purpose of elevation zones in stratified sampling?
A. To remove organisms
B. To decorate the mountain
C. To make counting difficult
D. To divide the study area into groups

10. Why do researchers collect samples instead of studying all organisms?
A. Because mountains are small
B. Because all organisms are the same
C. Because researchers do not need data
D. Because sampling is faster and more practical

ANG 3RD STATION NOW CLCIKABLE (PROGRESS CHART IS LIGHTED FOR STATION 1 AND 2)



31ST PLATE : TITLE : Green Growth Lab: The In vitro Propapagation of Moss (Kanang welcome ilisan aning naa sa kilid. PWEDE RA GAGMAYAN ANG CLICKABLE SUBPARTS). NAAY AVATAR NGA MAGSTORYA ANANG WELCOME YOUNG RESEARCHER……..



Welcome, young researchers! In this station, you will enter the Green Growth Laboratory, where scientists carefully grow moss inside sterile containers through a process called in vitro propagation.
But this station is not only about growing moss. More importantly, it will train you to think and work like real researchers by learning how to:

- **Plan an effective data collection procedure**
- **Collect accurate data using appropriate research instruments**

As you explore the fascinating process of moss propagation, you will also discover that successful scientific investigations depend on organized planning and reliable data gathering.

MAKE THE DISCOVERY STOP CLICKABLE

32nd Plate: CONTENT (place the content at the blank space) if possible unta sir kanang mag scroll lang within the same page ang content kay medyu taas2 siya. Tas if Makaya pabutangan nako ug voice over sir.Peru ang voiceover pwede sad siya mapalong in case nga nay studyante dili ganahan maminaw ug tingug kay mas gusto niya magbasa ra.



CONTENT:

The term **in vitro** comes from a Latin phrase meaning *"within glass."* In science, it refers to growing organisms inside a controlled laboratory environment using glass containers, bottles, petri dishes or any sterile containers. In vitro propagation is a technique where small moss parts such as spores, leaves, or fragments are grown under sterile conditions with the proper nutrients, moisture, and light. Through this method, scientists can large numbers of moss plants, study their growth and development, preserve rare species, and observe how moss responds to different environmental conditions. Unlike moss naturally on rocks or tree trunks, in vitro propagation allows researchers to carefully control environmental factors and observe changes more systematically.



culture of moss stem media, produce growing

In moss in vitro propagation, researchers do not simply observe moss growing inside culture containers. Instead, they gather scientific evidence to understand how different environmental factors affect moss growth and development. Researchers may investigate whether a nutrient solution promotes faster growth, whether light intensity affects moss color and size, or how long moss takes to develop under controlled laboratory conditions. To answer these research questions accurately, scientists must first plan a systematic data collection procedure and use appropriate instruments that will help them gather reliable observations and measurements throughout the investigation.

A **data collection procedure** serves as a guide for how information will be gathered throughout the investigation. Before starting the experiment, researchers must first decide what type of data should be collected. In moss propagation studies, the data may include moss height, number of new shoots, moisture levels, or growth rate over time. Researchers must also determine when the data should be collected. Some experiments require daily observations, while others may involve recording data every three days or once a week. Consistent timing is important because it helps ensure fairness and accuracy in the investigation.

After planning the data collection procedure, researchers must then select the most appropriate instruments for gathering accurate observations and measurements. The instruments used in the investigation depend on the type of data needed during the experiment. Using suitable research instruments helps improve the validity and reliability of the collected information, making the findings more accurate and dependable. Accurate data collection strengthens the quality of the research and helps scientists develop reliable conclusions about moss growth and development under controlled laboratory conditions.

Below are some common research instruments that can be used in this study:

Instrument	IMAGE	Purpose
Ruler		Measures moss height or spread
Thermometer		Measures temperature inside the laboratory
Hygrometer		Measures humidity level
Digital weighing scale		Measures biomass or mass

Camera or smartphone		Documents visual changes
Observation sheet		Records daily observations
Stopwatch or timer		Monitors time intervals
Graduated cylinder		Measures volume of liquid accurately
Microscope		Observes tiny moss structures

Remember, young researchers: successful investigations do not only depend on growing moss properly. They also depend on how carefully you collect, record, and analyze your data using the appropriate research instruments.

33rd PLATE: RESEARCH MISSION NOW CLICKABLE



PLATE 34: RESEACRH MISSION. ADD CLICKABLE BUTTON TO SATRT THE MAIN ACTIVITY



PLATE 35: Interactive Features

- Wrong placement shakes the item slightly
- Correct placement glows green



Correct Sequence

- Identify the data needed
- Determine the schedule of observation
- Select the appropriate instruments
- Observe the moss cultures
- Record observations in a data sheet
- Organize and analyze the gathered data

PLATE 36: ACTIVITY : PART 2

Interactive Features

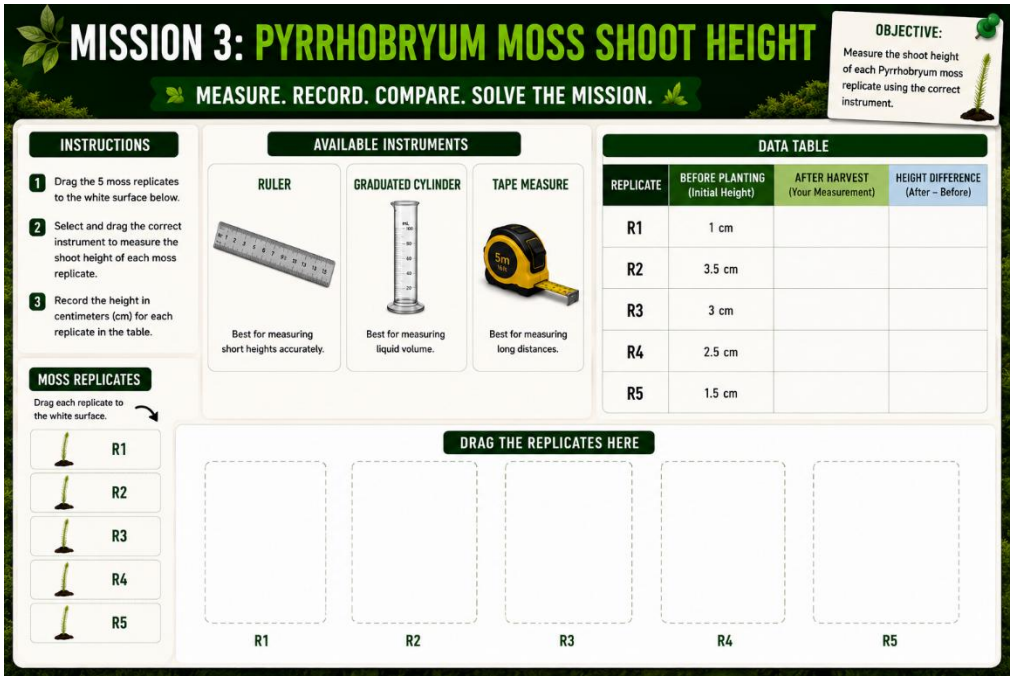
- Instruments appear as clickable icons
- Correct answers produce a glowing effect
- Wrong answers produce a soft warning sound



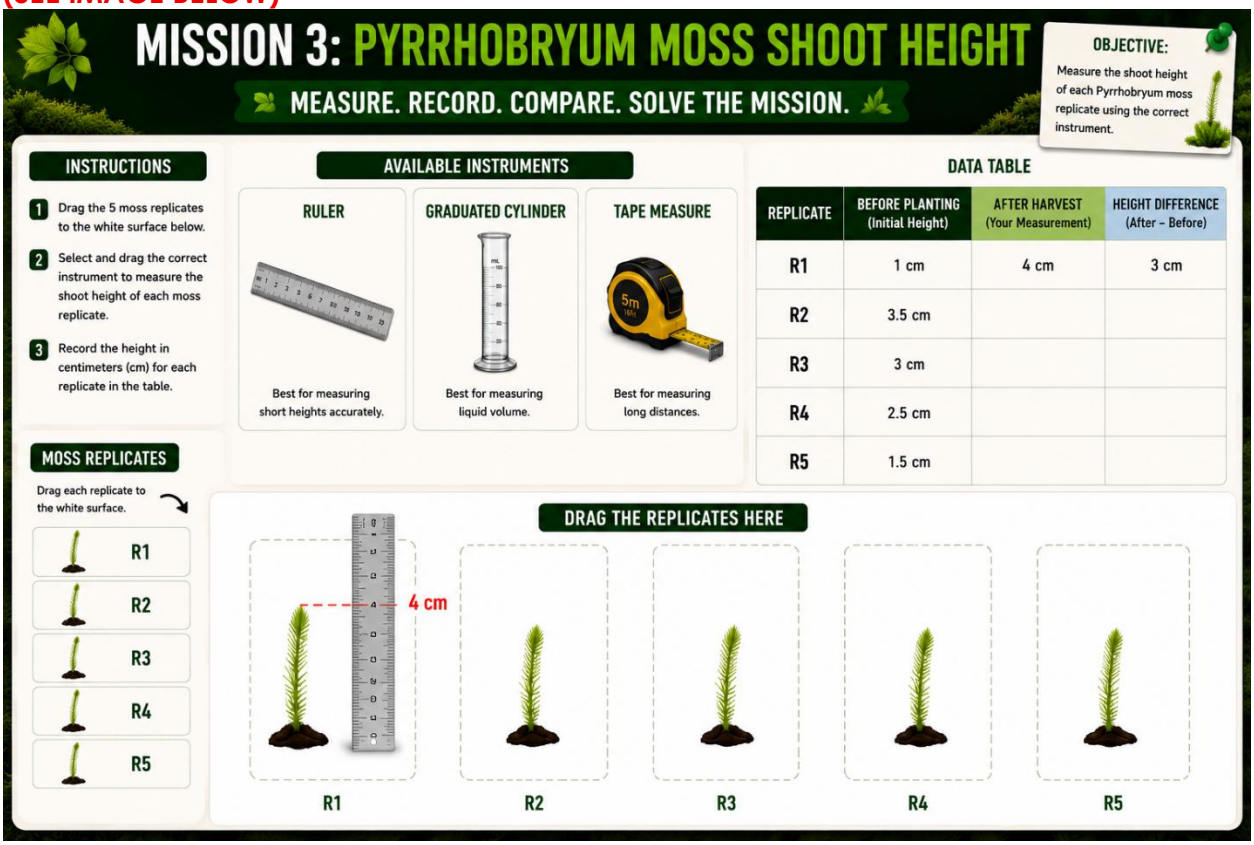
Answer

1. Ruler
2. Digital balance
3. Microscope
4. Digital thermometer
5. Record note
6. Hygrometer
7. Camera
8. Graduated cylinder

PLATE 35: **FURTHER INSTRUCTIONS FOR THIS ACTIVITY IS FOUND BELOW:**



(SIR SA KANI NA ACTIVITY DILI KO SURE IF MAKAYA NEMU SIYA PERU I HOPE MAKAYA. E DRAG SA BATA ANG 5 KA PLANT DRAWING PADULONG SA BLANK SHEETS. LAHI LAHI SIYA UG HEIGHT KANA PLANT INING MAPLASTAR NA SA BATA (R1- 4; CM; R2- 5 CM, R3-5.5 CM, R4-3.5 CM, R5- 4.5 CM) AND THEN AFTER SA PLANT ISUNOD NILA UG DRAG ANG RULER PARA ISUKOD SA PLANT UG DAPAT KATO NGA HEIGHT ILA ISULAT SA TABLE UG DILI MAO ILANG ISULAT DILI SIYA DAWATON SA TABLE. ENGUN ANI BALE (SEE IMAGE BELOW)



(DILI NI SIYA INCLUDED AS PLATE SIR MAO LANG NI ANG CONCEPT NA BUHATON SA BATA ANI NA ACTIVITY. DAPAT SILA PUD MAGCOUNT GAMIT ANG RULER)

PLATE 36. Takeaway QUESTION. STUDENTS WILL WRITE THEIR ANSWER IN A BOX. ADD CLICKABLE BUTTON NA BAKALAIK SILA SA NEXT PLATE

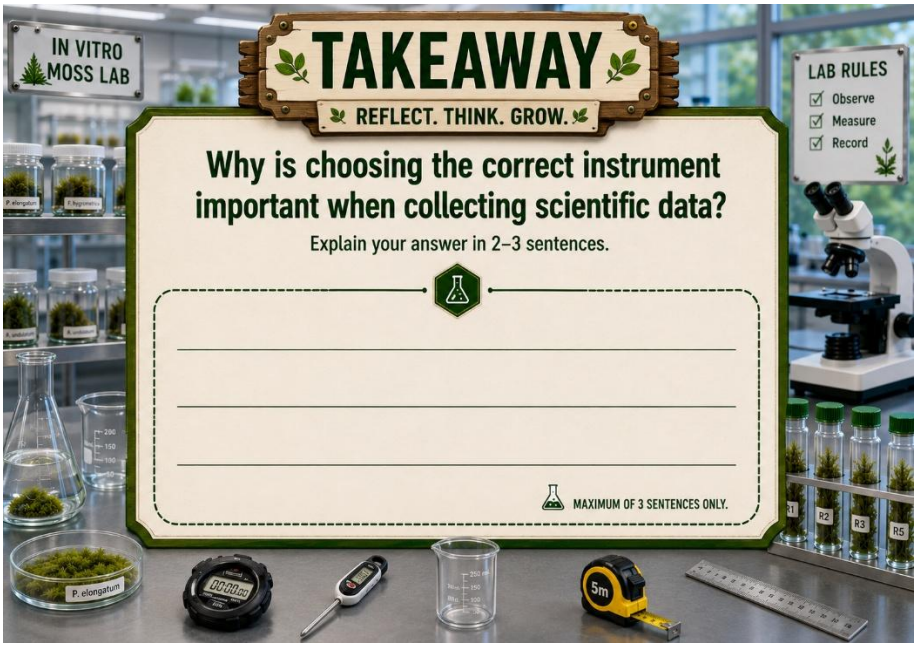


PLATE 37:
MAKE THE TRAIL CHECKPOINT CLICKABLE



MUGAWAS DAYUN ANG MAIN ASSESSMENT



QUESTIONS:

1. A student needs to measure the shoot height of a moss sample accurately. Which instrument should be used?

A. Ruler ✅
B. Camera
C. Beaker
D. Stopwatch

---

2. During the investigation, the researcher wants to compare the mass of moss samples before and after harvest. Which instrument is most appropriate?

A. Thermometer
B. Tape Measure
C. digital caliper
D. Digital Balance ✅

---

3. A learner wants to observe very small moss structures that cannot be seen clearly by the naked eye. Which instrument should be used?

A. Beaker
B. Hygrometer
C. Stopwatch
D. Microscope ✅

---

4. Why is choosing the correct instrument important in scientific investigations?

A. To finish slowly
B. To avoid observations
C. To make the laboratory colorful
D. To collect accurate scientific data ✅

---

5. A researcher wants to compare the physical appearance of moss cultures every week. Which tool would provide the best visual record?

A. Ruler
B. Beaker
C. Camera ✅
D. Stopwatch

6. A group collected measurements using different instruments incorrectly. What is the most likely effect on their investigation?
A. Faster moss growth
B. More colorful cultures
C. Less reliable research data ✅
D. More laboratory equipment

Analyze the laboratory data carefully and answer the questions that follow.

REPLICATE	BEFORE PLANTING	AFTER HARVEST	HEIGHT DIFFERENCE
R1	1.0 cm	4.0 cm	Students ang mag input
R2	3.5 cm	6.5 cm	Students ang mag input
R3	3.0 cm	5.0 cm	Students ang mag input
R4	2.5 cm	6.0 cm	Students ang mag input
R5	1.5 cm	4.5 cm	Students ang mag input

7. Which replicate showed the greatest increase in shoot height?
A. R1
B. R3
C. R4 ✅
D. R5

8. Which replicate had the tallest moss after harvest?
A. R2 ✅
B. R3
C. R4
D. R5

9. Based on the table, what can be concluded about the moss samples after one month?
A. All moss samples died
B. Moss samples showed no changes
C. Most moss samples increased in height ✅
D. Moss samples became shorter and bigger

10.. Why is organizing collected data in tables important during scientific investigations?
A. To analyze data easily
B. To make the report longer
C. To avoid laboratory work
D. To make the table colorful

PLATE 38: BALIK SA EXPIDITION MAP PARA MACLICK ANG 4TH STATION. LIGHTED NAG PROGRESS BAR UP TO 3RD STATION

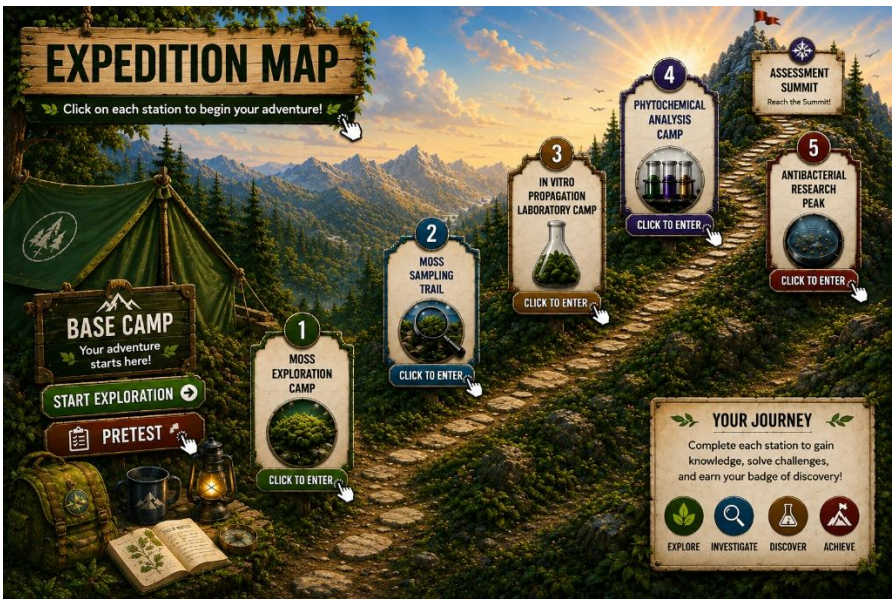


PLATE 39: TITLE: Bioactive Secrets: The Phytochemical Screening of Moss (Kanang welcome ilisan aning naa sa kilid. PWEDE RA GAGMAYAN ANG CLICKABLE SUBPARTS)

NAAY AVATAR NGA MAGSTORYA ANANG WELCOME YOUNG RESEARCHER……..



MAKE DISCOVERY STOP CLICKABLE

Welcome, young researchers! In this station, you will enter the Bioactive Discovery Laboratory, where scientists investigate the hidden chemical compounds found inside moss through a process called phytochemical screening.

But this station is not only about identifying bioactive compounds. More importantly, it will train you to think and work like real researchers by learning how to:

- Organize and present scientific data using tables and appropriate graphs
- Interpret data based on observed patterns and results

As you explore the fascinating bioactive secrets of mosses, you will also discover that scientific investigations become meaningful only when data are properly presented and carefully interpreted.

PLATE 40. CONTENT PROPER

CONTENT (place the content at the blank space) if possible unta sir kanang mag scroll lang within the same page ang content kay medyu taas2 siya. Tas if Makaya pabutangan nako ug voice over sir.Peru ang voiceover pwede sad siya mapalong in case nga nay studyante dili ganahan maminaw ug tingug kay mas gusto niya magbasa ra.



CONTENT PROPER

Mosses may appear small and simple, but recent studies have revealed that they contain important bioactive compounds that help them survive in different environmental conditions. These naturally occurring substances help protect mosses from harmful microorganisms, environmental stress, and predators. Some compounds may also possess antioxidant, antimicrobial, and medicinal properties, making mosses valuable in scientific and biological research.

To find out and study these compounds, scientists perform a laboratory process called phytochemical screening. **Phytochemical profiling** is used to identify and study the naturally occurring chemical compounds present in plants, including mosses. Through this laboratory process, scientists are able to examine the different bioactive compounds found in moss samples and better understand their possible functions, ecological importance, and potential applications in science and medicine.

Below are some common phytochemicals found in mosses and their functions:

Phytochemical	Function
Alkaloids	Help protect mosses against microorganisms and predators
Flavonoids	Function as antioxidants and protect cells from environmental stress
Tannins	Provide defense against pathogens and herbivores
Saponins	Possess antimicrobial properties
Terpenoids	Help mosses adapt to environmental stress
Phenolic Compounds	Help protect cells from oxidative damage

(Information adapted from *Biologically Active Compounds from Bryophytes* by Y. Asakawa (2007); *Bryophyte Ecology: Chemical Defense and Secondary Metabolites* by J. M. Glime (2017); and *Secondary Metabolites in Bryophytes: An Ecological Aspect* by C. F. Xie et al. (2009).)

In phytochemical studies of mosses, researchers gather scientific evidence to find out, analyze, interpret and compare the bioactive compounds present in different moss species and evaluate patterns recorded during the investigation. And, to communicate these findings clearly and accurately, scientists organize the collected observations using appropriate tabular and graphical forms.

One common way of presenting scientific data is through tables. Tables organize information into rows and columns, making it easier for researchers to compare observations among moss samples. In phytochemical screening studies, tables may show the presence or absence of compounds in different moss species. Organized tables help simplify large amounts of information and allow researchers to identify similarities and differences more clearly. Aside from tables, researchers also use

graphs to visually present research findings. Graphs help researchers observe patterns, trends, and relationships in the gathered data.

Below are some common tabular and graphical forms used in phytochemical screening studies:

TABULAR/GRAPHICAL FORMS USED IN RESEARCH

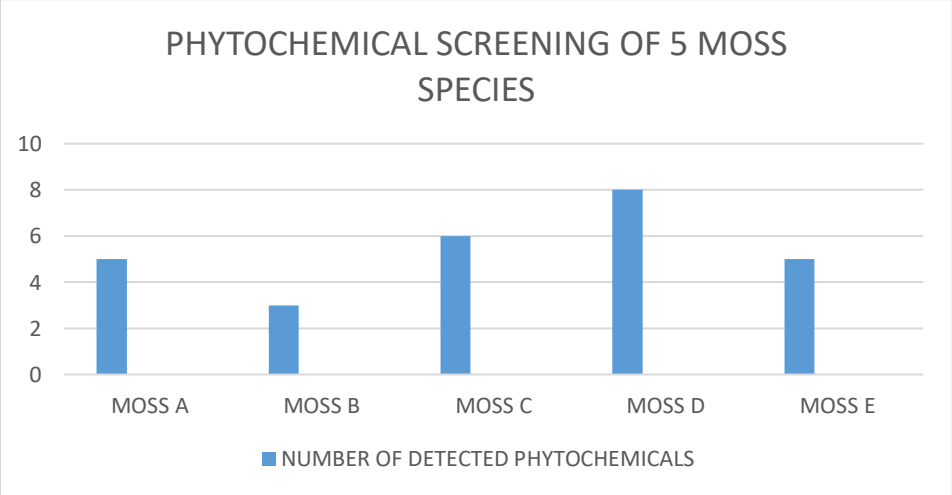
Tabular/Graphical Form	Purpose
Data Table	Organizes and summarizes data into rows and columns for easier analysis.
Bar Graph	Compares data among different categories or groups.
Pie Graph	Shows proportions or percentages of parts within a whole.
Line Graph	Presents changes, trends, or patterns over time.
Frequency Table	Summarizes the number of times observations or responses occur.

Data interpretation is another important skill in scientific investigations. Interpreting data means explaining the meaning of the results based on the information presented in tables and graphs. Researchers analyze patterns, identify relationships, and draw conclusions supported by evidence gathered during the investigation. For example, if one moss species shows a higher number of detected phytochemicals than another, scientists may conclude that the species has greater bioactive potential. Proper interpretation helps researchers communicate scientific findings accurately and meaningfully.

Let us say, for example, that researchers conducted phytochemical screening on five moss species and recorded the number of detected phytochemicals found in each sample.

Moss Species	Number of Detected Phytochemicals
Moss A	5
Moss B	3
Moss C	6
Moss D	8
Moss E	5

The data above may be presented using a **bar graph** to compare the number of phytochemicals detected in each moss species. Through the graph, researchers can easily identify which sample contains the highest or lowest number of detected compounds.



Interpretation: The bar graph shows the number of detected phytochemicals present in five moss species. Among the samples, Moss D has the highest number of detected phytochemicals with a total of 8 compounds while Moss B has the lowest number with only 3 compounds detected. Moss C has 6 detected phytochemicals, whereas Moss A and Moss E both have 5. This shows that the phytochemical content varies among the moss species, with Moss D showing the richest phytochemical composition among the five samples.

From the example given above, it is evident that tables and graphs help researchers organize data systematically, compare results easily, and interpret scientific findings more clearly and accurately.

Remember, young researchers, successful scientific investigations depend on how carefully data are organized, presented, and interpreted using appropriate tabular and graphical forms. Through accurate data presentation and interpretation, researchers are able to communicate findings clearly, identify meaningful patterns, and draw reliable scientific conclusions based on evidence gathered during the research investigation.

PLATE 41: MAKE THE RESEARCH MISSION CLICKABLE



PLATE 41: ADD BUTTON TO START THE ACTIVITY



PLATE 42. DAPAT NAAY CLICKABLE ICON PARA SA NEXT ACTIVITY



ACTIVITY 1: (DRAG AND DROP)

🟩 1

This graphical form is used to compare data among different categories or groups.

✅ Correct Answer: 📊 Bar Graph

🟩 2

This graphical form shows proportions or percentages of parts within a whole.

✅ Correct Answer: 🥧 Pie Graph

🟩 3

A scientist records the increase in detected phytochemicals in a moss culture over five weeks and wants to show the changes over time.

✅ Correct Answer: 📈 Line Graph

🟩 4

A researcher wants to summarize how many times specific observations appeared during the investigation using tally marks and frequencies.

✅ Correct Answer: 📑 Frequency Table

🟩 5

A researcher wants to organize laboratory observations systematically before making graphs and interpretations.

✅ Correct Answer: 📋 Data Table

PLATE 43.



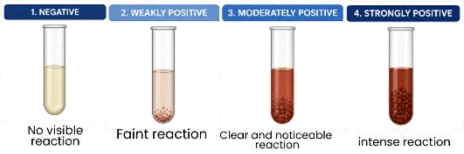
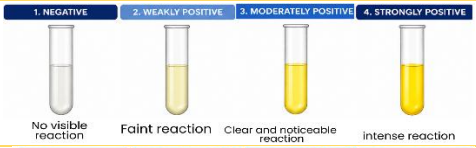
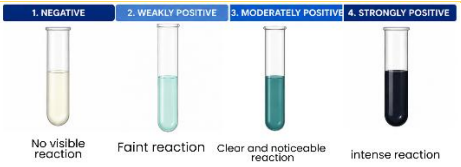
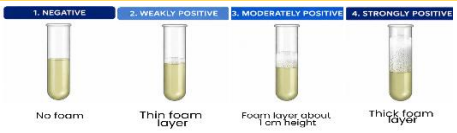
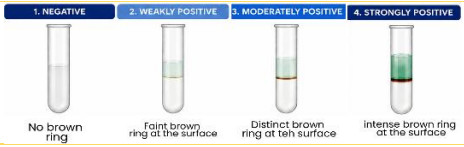
ACTIVITY OVERVIEW

In this activity, you will analyze the phytochemical screening results obtained from two moss samples, Genus *Pyrrhobryum* and Genus *Ectropothecium*. You will carefully observe the laboratory reactions produced during different phytochemical tests and determine whether the compounds are negative (–), weakly present (+), moderately present (++), or strongly present (+++).

Using laboratory observations such as foam formation, color changes, and precipitate formation, you will analyze the intensity of the reactions and record your findings in a scientific results table just like real researchers do during phytochemical investigation.

Before starting the activity, let us have an overview first on how scientists analyze the phytochemicals present in every plant sample. Scientists use different laboratory tests to identify the presence of bioactive compounds in plant samples. When a reagent is added to a moss extract, observable reactions may occur such as: formation of foam, color changes, precipitates, or visible layers. The strength of the reaction helps determine the level of phytochemicals present in the sample.

GENERAL GUIDELINES FOR INTERPRETING RESULTS		
Symbol	**Meaning**	**Laboratory Observation**
–	Negative	No visible reaction
+	Weakly Positive	Weak or faint reaction observed
++	Moderately Positive	Clear and noticeable reaction
+++	Strongly Positive	Very intense reaction observed

COMMON PHYTOCHEMICAL TESTS			
Phytochemical	**Sample Test Used**	**Possible Reaction if Present**	**Illustration**
Alkaloids	Wagner's Test	Reddish-brown precipitate forms	
Flavonoids	Alkaline Reagent Test	Yellow coloration develops	
Tannins	Ferric Chloride Test	Blue-black or dark coloration appears	
Saponins	Foam Test	Stable foam forms after shaking	
Glycosides	Keller-Killiani Test	Brown ring forms between the layers	

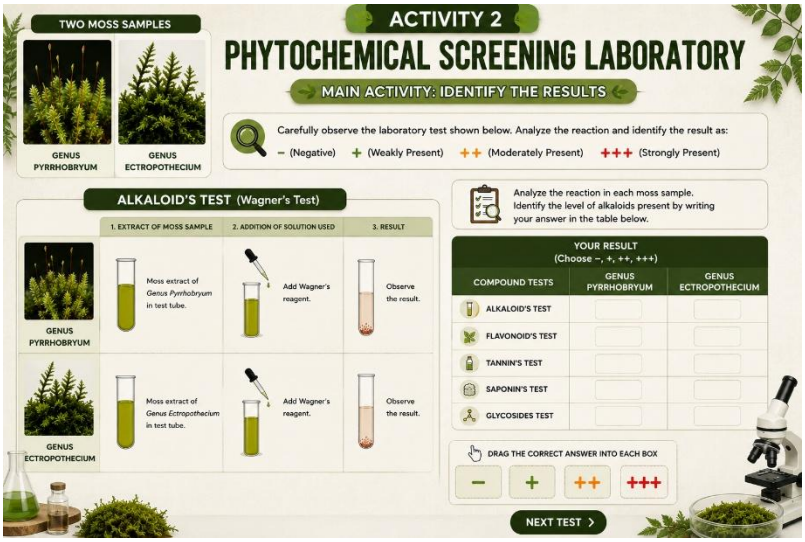
(Remark: The tests presented in the table are only some of the commonly used phytochemical tests. Other reagents and testing methods may also be used to detect the presence of these phytochemicals depending on the laboratory procedure and research design).

INSTRUCTIONS:

1. Carefully observe the laboratory reactions shown for each moss sample.

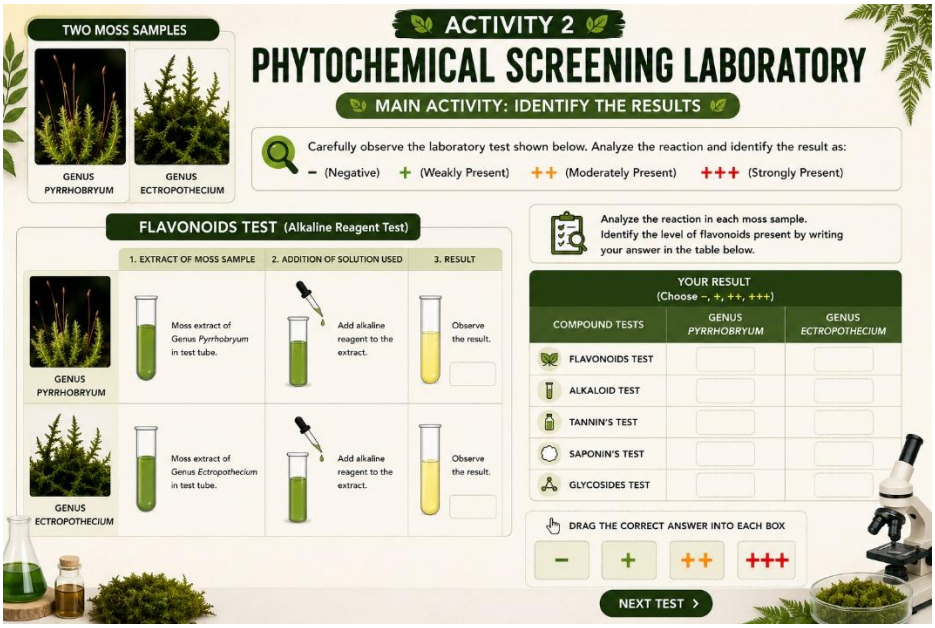
2. Analyze the intensity of the reaction based on the guide provided.
3. Determine whether the phytochemical is:
    - Negative (–)
    - Weakly Positive (+)
    - Moderately Positive (++)
    - Strongly Positive (+++)
4. Record your observations in the scientific results table.
5. Compare the phytochemical contents of the two moss samples.
6. Answer the interpretation guide questions after completing the table.

PLATE 44: DRAGGABLE ANG (-,+,++,+++) (SIR PLEASE NOTE THAT THE ILLUSTRATIONS HERE IS CRUCIAL). DILI DAWATON SA BOX KUNG ILA ANSWER IS WRONG



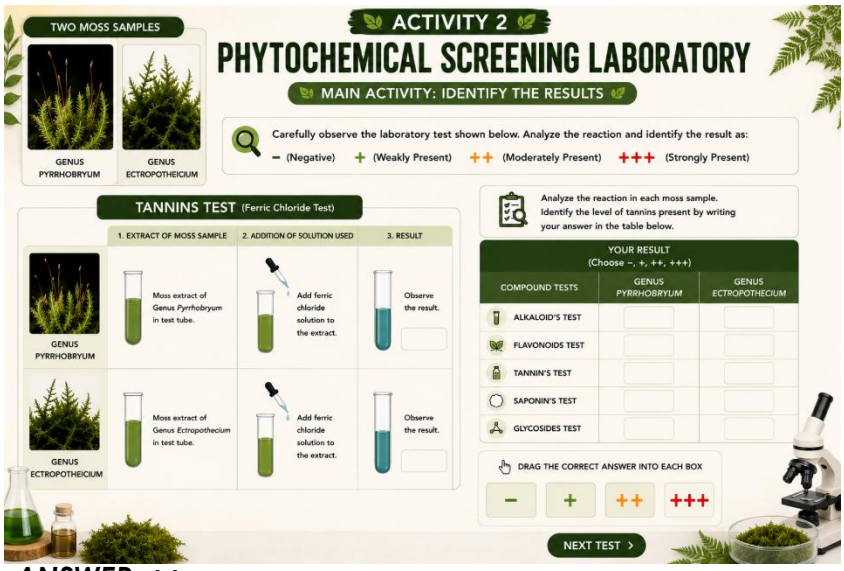
ANSWER: +

PLATE 45: DRAGGABLE ANG (-,+,++,+++) (SIR PLEASE NOTE THAT THE ILLUSTRATIONS HERE ARE CRUCIAL). DILI DAWATON SA BOX KUNG ILA ANSWER IS WRONG



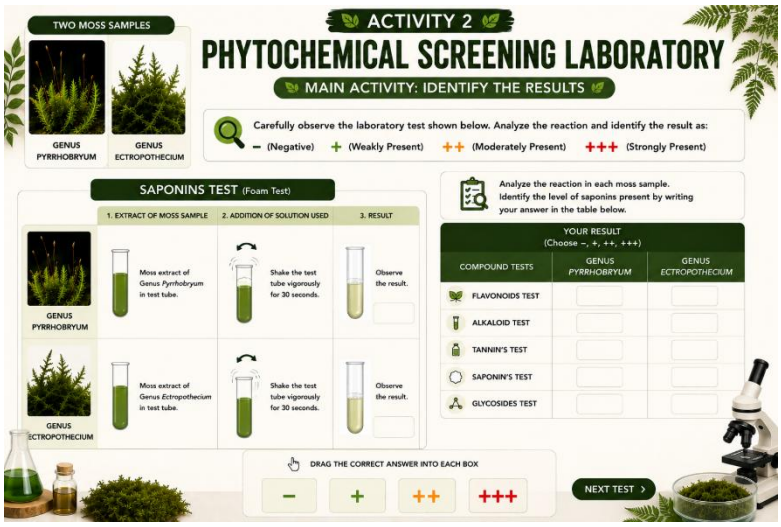
ANSWER: ++

PLATE 46: DRAGGABLE ANG (-,+,++,+++) (SIR PLEASE NOTE THAT THE ILLUSTRATIONS HERE ARE CRUCIAL). DILI DAWATON SA BOX KUNG ILA ANSWER IS WRONG



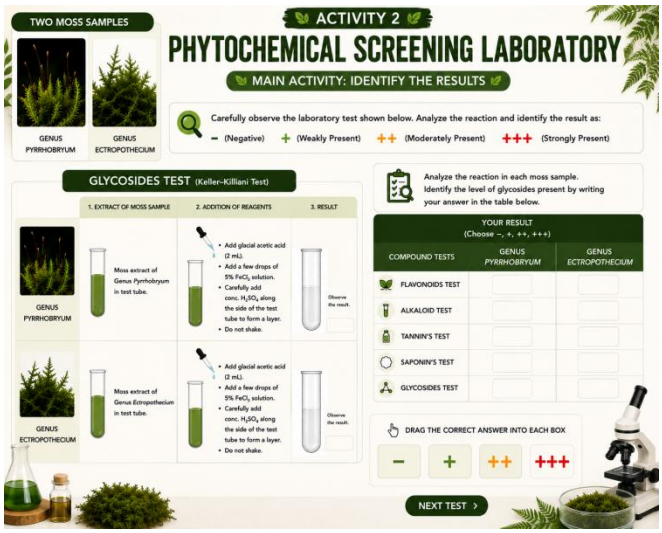
ANSWER: ++

PLATE 47: DRAGGABLE ANG (-,+,++,+++) ((SIR PLEASE NOTE THAT THE ILLUSTRATIONS HERE IS CRUCIAL). DILI DAWATON SA BOX KUNG ILA ANSWER IS WRONG



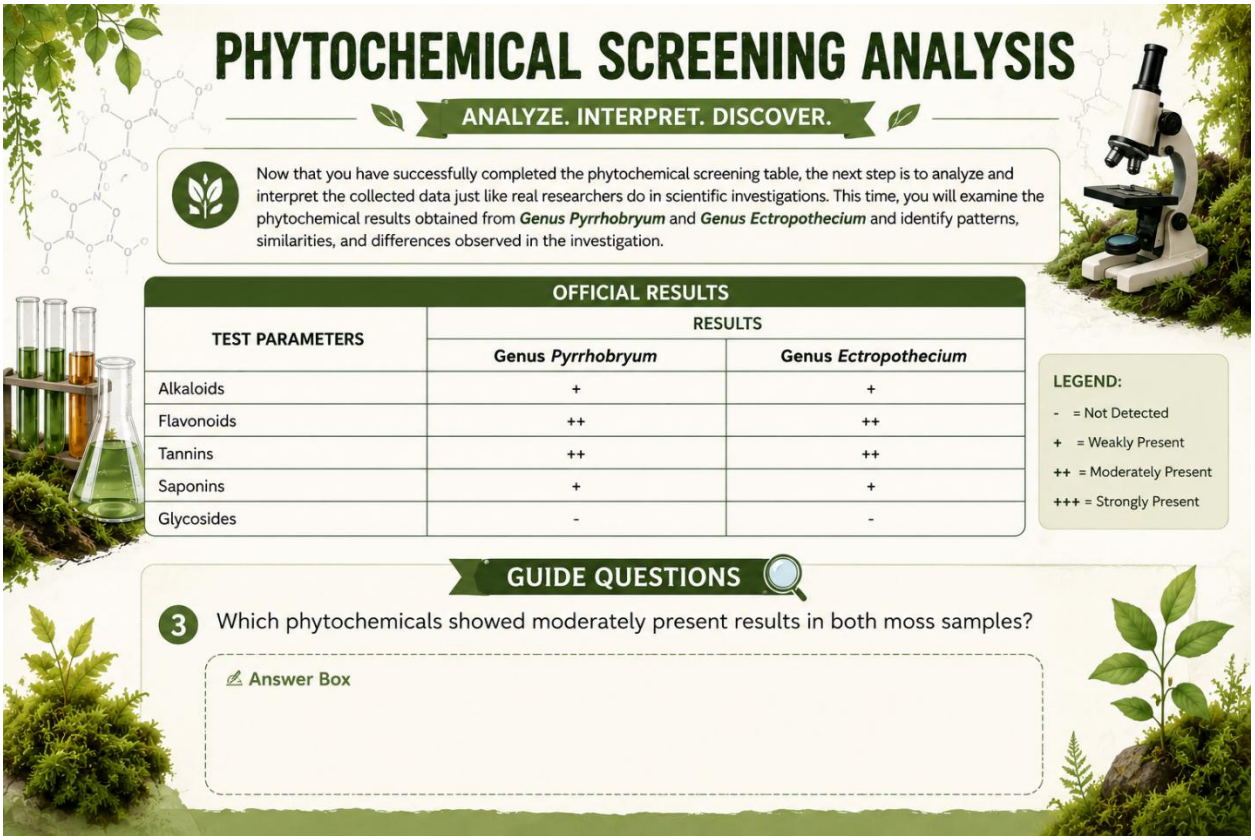
ANSWER: +

PLATE 48: DRAGGABLE ANG (-,+,++,+++) (SIR PLEASE NOTE THAT THE ILLUSTRATIONS HERE ARE CRUCIAL). DILI DAWATON SA BOX KUNG ILA ANSWER IS WRONG



ANSWER: -

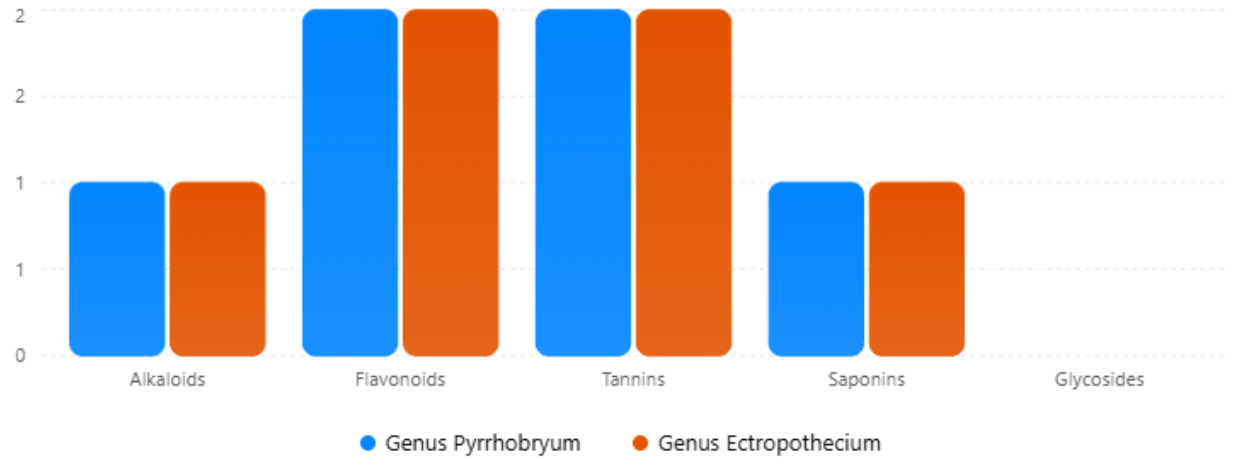
PLATE 49: CONTINUATION SA PREVIOUS ACTIVITY BUT THIS TIME E INTERPRET NA NILA 9CHANGE QUESTION NUMBER TO 1) THEN CONTINUE WITH THE REMIANING QUESTIONS LISTED BELOW:



🟩 **Question 1** (sir possible kaya na sila jud ang magconstruct sa bar graph? Basin pwede naay graph daan tas edrag and drop lang nila ang lain2 na sizes sa bars for each species. Or any way lang sir na sila unta makadraw sa graph)

Construct a bar graph based on the phytochemical screening results of *Genus Pyrrhobryum* and *Genus Ectropothecium*. Use the following numerical values for the symbols: (- = 0, + = 1, ++ = 2, +++ = 3). Use blue bar for *Genus Pyrrhobryum* and orange bar for *Genus Ectropothecium*.

Answer:



🟩 Question 2

Which phytochemicals showed moderately present results in both moss samples?

✍️ Answer : Flavonoids and tannins

🟩 Question 3

Which phytochemical compound was absent in both Genus *Pyrrhobryum* and Genus *Ectropothecium*?

✍️ Answer : Glycosides

🟩 **Question 4**
Which phytochemicals were only weakly present in both moss samples?
✍️ Answer: alkaloid and saponin

🟩 **Question 5**
What can you conclude about the phytochemical composition of the two moss species based on the results?
✍️ Answer: Based on the results, both *Genus Pyrrhobryum* and *Genus Ectropothecium* have very similar phytochemical compositions because they showed the same level of presence for all tested compounds. Flavonoids and tannins were moderately present (++), indicating that these compounds may be abundant in both moss species. Alkaloids and saponins were only weakly present (+), while glycosides were absent (-) in both samples. These similarities suggest that the two moss species may possess comparable chemical properties and potential biological activities.

PLATE 5O: TAKEAWAY QUESTION:
Why is it important to analyze and interpret phytochemical screening results using tables and graphs in scientific investigations?

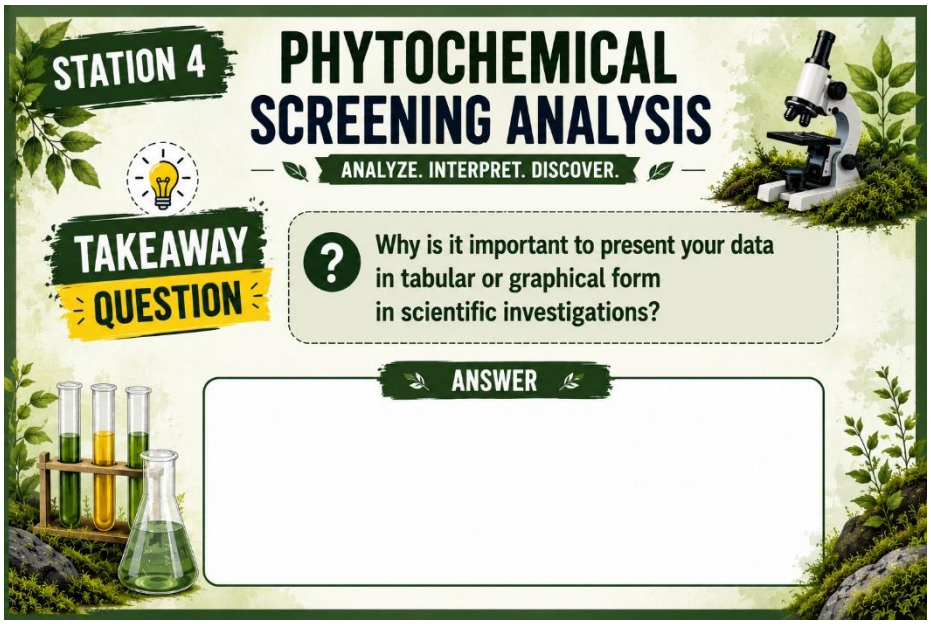
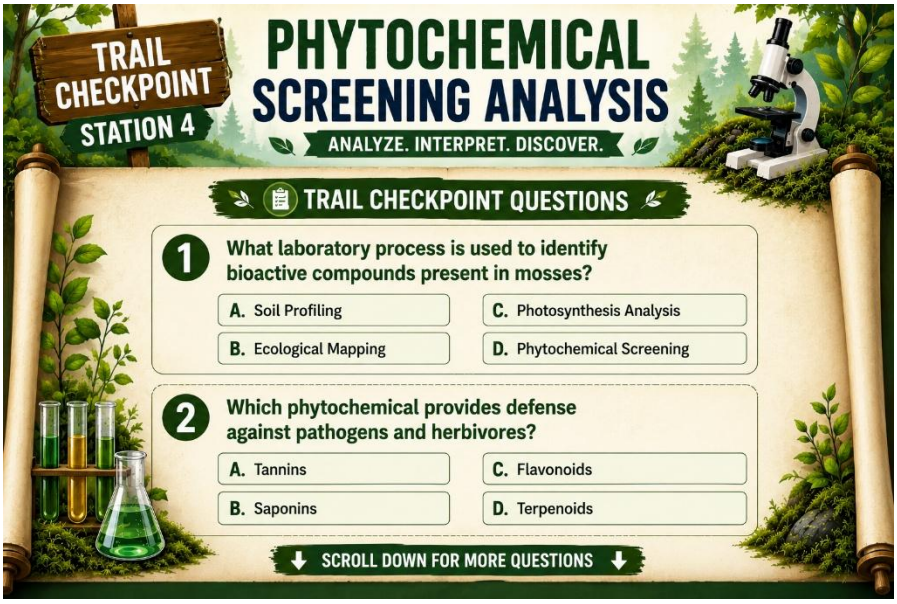


PLATE 51. BALIK ANI NA SLIDE, MAKE TEH TRAIL CHECKPOINT CLICKABLE



PLATE 52:



💉 Multiple Choice Questions

1. What laboratory process is used to identify bioactive compounds present in mosses?

A. Soil Profiling
B. Ecological Mapping
C. Photosynthesis Analysis
D. Phytochemical Screening

✅ **Answer:** D. Phytochemical Screening

---

2. Which type of graph is BEST used to compare phytochemical results between two moss species?

A. Pie Graph
B. Line Graph
C. Scatter Plot
D. Bar Graph

✅ **Answer:** D. Bar Graph

---

3. Which type of graph is MOST suitable for showing changes or trends in data over time?
A. Pie Graph
B. Bar Graph
C. Line Graph
D. Pictograph
✅ **Answer:** C. Line Graph

---

4. Why do scientists organize data into tables?
A. To avoid interpreting results
B. To decorate research papers
C. To make investigations longer
D. To organize information clearly
✅ **Answer:** D. To organize information clearly

---

5. What is the main purpose of interpreting scientific data?
A. To make graphs colorful
B. To shorten the investigation
C. To repeat experiments only
D. To explain the meaning of the results
✅ **Answer:** D. To explain the meaning of the results

📊 **Use the table below to answer Questions 9–15.**

Phytochemical	**Genus *Pyrrhobryum***	**Genus *Ectropothecium***
Alkaloids	+	+
Flavonoids	++	++
Tannins	++	++
Saponins	+	+
Glycosides	-	-

Legend:
- = Absent
- = Weakly Present
  ++ = Moderately Present
  +++ = Strongly Present

---

6. Which phytochemicals showed moderately present results in both moss samples?
A. Alkaloids and Saponins
B. Alkaloids and Glycosides
C. Glycosides and Tannins
D. Flavonoids and Tannins
✅ **Answer:** D. Flavonoids and Tannins

---

7. Which phytochemical compound was absent in both moss samples?
A. Saponins
B. Alkaloids
C. Glycosides
D. Flavonoids
✅ **Answer:** C. Glycosides

---

8. Which phytochemicals were weakly present in both moss samples?
A. Alkaloids and Saponins
B. Flavonoids and Glycosides

C. Glycosides and Tannins
D. Flavonoids and Tannins
✅ **Answer:** A. Alkaloids and Saponins

---

9. What can be concluded from the phytochemical results of the two moss species?
A. Only one moss species contains bioactive compounds.
B. The moss species have completely different phytochemical compositions.
C. Both moss species showed similar phytochemical compositions.
D. Glycosides were strongly present in both samples.
✅ **Answer:** C. Both moss species showed similar phytochemical compositions.

---

10. Why is it important to present scientific data in tabular or graphical form?
A. To make the investigation more difficult
B. To avoid drawing conclusions from evidence
C. To help organize, compare, and interpret data clearly
D. To confuse the readers with more information and data
✅ **Answer:** c. To help organize, compare, and interpret data clearly

5TH STATION
PLATE 53: BALIK EXPIDITION MAP PARA MACLICK ANG STATION 5

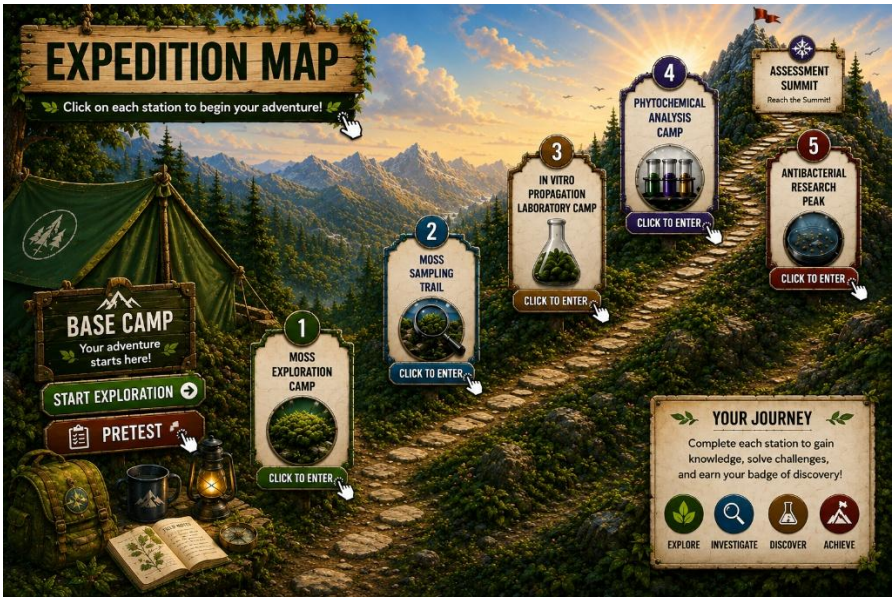


PLATE 54: MAKE THE DISCOVERY STOP CLICKABLE. NAAY AVATAR MAGSTORYA PARA SA WELCOME YOUNG... PWEDE RA gagmay ang



MACLICK AND DISCOVERY STOP

Welcome, young researchers! In this station, you will enter the Defense Laboratory, where scientists investigate the antibacterial properties of moss through a process called antibacterial assay. Here, you will discover that mosses produce natural compounds which helps protect them from harmful bacteria in their environment.

But this station is not only about testing moss extracts against bacteria. More importantly, it will train you to think and work like real researchers by learning how to:

• Plan data analysis using statistics and hypothesis testing

As you explore the fascinating antibacterial properties of mosses, you will also discover that successful scientific investigations require proper data analysis, careful interpretation of results, and logical decision-making based on scientific evidence.

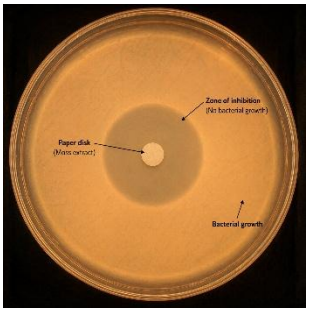
PLATE 55: DIRI IBUTANG ANG CONTENT SA LESSON 5



CONTENT

Mosses are small plants that usually grow in moist and shaded places where bacteria and other microorganisms are commonly present. Because mosses are constantly exposed to these microorganisms, some species produce natural compounds that help protect them from harmful bacteria and allow them to survive in their environment. These compounds may have antibacterial properties that can inhibit or prevent bacterial growth. Due to this ability, scientists study mosses to determine whether their extracts can be used as possible natural antibacterial agents.

One commonly used method in studying antibacterial activity is the Kirby-Bauer Method also known as disk diffusion method. In this procedure, paper disks soaked with moss extract are placed on agar plates containing bacterial cultures. After incubation, a clear area surrounding the disk called the zone of inhibition may appear. This clear zone shows that the bacteria were unable to grow near the moss extract because of its antibacterial effect. Generally, larger zones of inhibition show stronger antibacterial activity.



In scientific investigations like this, researchers do not rely on observations alone rather they carefully organize, analyze, and interpret the gathered data to make evidence-based conclusions. So, to ensure that the gathered data is properly analyzed and interpreted, researchers follow several important steps in planning data analysis using statistics and hypothesis testing.

Here are the steps researchers follow in planning data analysis using statistics and hypothesis testing:

To better understand these steps, let us refer to a sample antibacterial investigation created for the purpose of helping us understand how data analysis is planned in an experiment. In this example, the moss genera Pyrrhobryum and Ectropothecium were tested against Escherichia coli using the Kirby-Bauer Method. The investigation also included a positive control containing a known antibacterial agent and a negative control containing a substance without antibacterial properties. To make the results more reliable, five replicates were conducted for each treatment. After incubation, the zones of inhibition produced by the moss extracts and controls were measured and recorded.

Step 1 – Identify the Data to be Gathered

In this step, the researcher identifies the information needed from the investigation. This helps the researcher determine the measurements that are important in answering the research problem.

Example:

The researcher measures and records the diameters of the zones of inhibition produced by Pyrrhobryum, Ectropothecium, and the controls against Escherichia coli for all five replicates.

Step 2 – Formulate the Hypothesis

In this step, the researcher formulates a hypothesis or scientific prediction about the possible outcome of the investigation. A hypothesis is important because it guides the researcher in analyzing and interpreting the gathered data.

Example:

Null Hypothesis (H_0): *There is no significant difference between the antibacterial effects of the moss genera Pyrrhobryum and Ectropothecium against Escherichia coli.*

Alternative Hypothesis (H_1): *There is a significant difference between the antibacterial effects of the moss genera Pyrrhobryum and Ectropothecium against Escherichia coli.*

Step 3 – Organize the Gathered Data
After collecting the measurements, the researcher organizes the data into tables for easier comparison and interpretation. Organizing the data helps the researcher identify patterns and differences among treatments.

Example:

Treatment	*Replicate 1*	*Replicate 2*	*Replicate 3*	*Replicate 4*	*Replicate 5*
Pyrrhobryum	*18 mm*	*17 mm*	*19 mm*	*18 mm*	*18 mm*
Ectropothecium	*12 mm*	*11 mm*	*13 mm*	*12 mm*	*12 mm*
Positive Control	*22 mm*	*21 mm*	*22 mm*	*23 mm*	*22 mm*
Negative Control	*0 mm*	*0 mm*	*0 mm*	*0 mm*	*0 mm*

*(**Note:** The data presented above are hypothetical or sample data created only for the purpose of demonstrating how data may be organized and analyzed in an antibacterial investigation).*

Step 4 – Choose the Appropriate Statistical Tool
In this step, the researcher selects statistical tools that can help analyze and present the gathered data clearly and accurately. Statistical tools are important because they help summarize results, compare treatments, and determine whether the observed differences are significant.

One commonly used statistical tool in experimental research is the mean or average. Researchers usually use the mean because it helps summarize repeated measurements into a single value that represents the overall result of the treatment. This makes the gathered data easier to analyze and compare.

Other statistical tools commonly used in experimental research include percentage, t-test, and Analysis of Variance (ANOVA). Graphical presentations such as bar graphs may also be used to visually compare the gathered data.

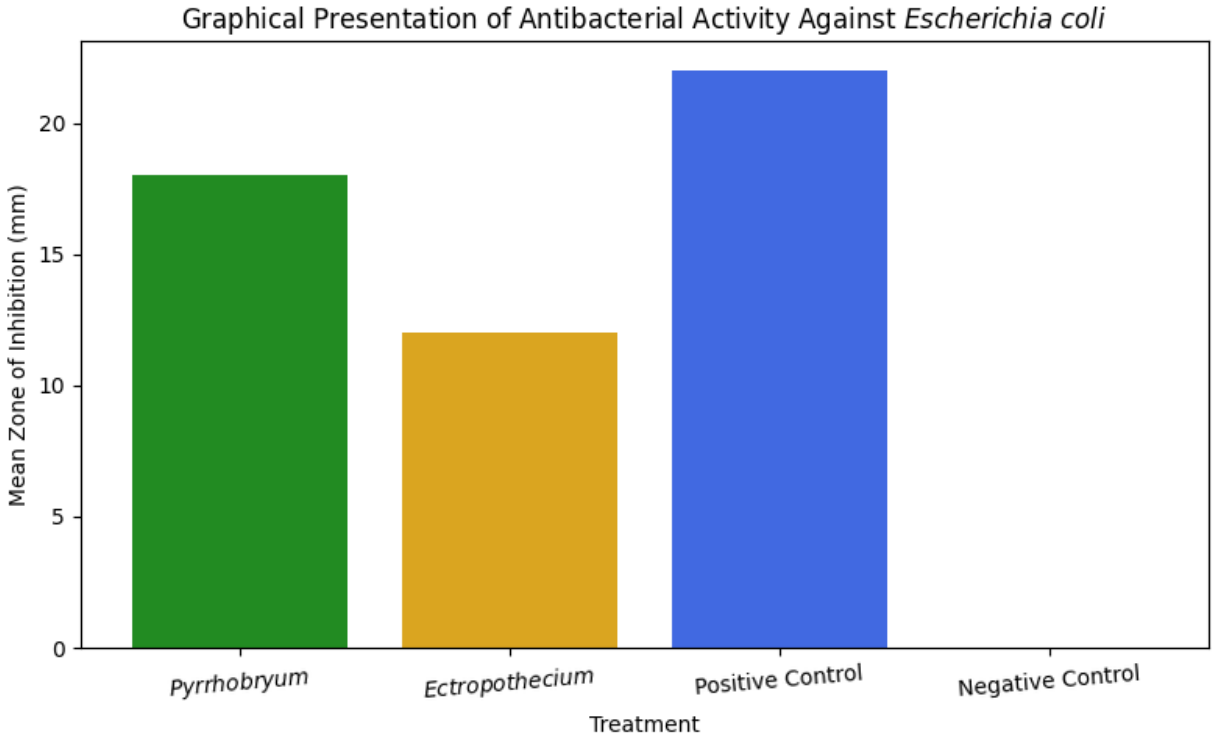
Example:
In this investigation, the researcher computes the mean zone of inhibition from the five replicates for each treatment to determine the antibacterial activity of Pyrrhobryum and Ectropothecium against Escherichia coli.

Treatment	*Replicate 1*	*Replicate 2*	*Replicate 3*	*Replicate 4*	*Replicate 5*	***Mean (mm)***
Pyrrhobryum	*18 mm*	*17 mm*	*19 mm*	*18 mm*	*18 mm*	*18.0 mm*
Ectropothecium	*12 mm*	*11 mm*	*13 mm*	*12 mm*	*12 mm*	*12.0 mm*
Positive Control	*22 mm*	*21 mm*	*22 mm*	*23 mm*	*22 mm*	*22.0 mm*
Negative Control	*0 mm*	*0 mm*	*0 mm*	*0 mm*	*0 mm*	*0 mm*

For example, the mean for Pyrrhobryum is computed as:

$$Mean = \frac{18 + 17 + 19 + 18 + 18}{5} = 18.0\ mm$$

The computed means may then be presented using a bar graph to visually compare the antibacterial effects of the treatments. Since the investigation compares the antibacterial activities of two moss genera, a t-test may also be used to determine whether the difference between their antibacterial effects is significant.



Step 5 – Compare and Interpret the Results

In this step, the researcher examines and interprets the gathered results to determine which moss genus showed stronger antibacterial activity. This helps the researcher explain the meaning of the findings based on the evidence gathered from the investigation.

Example:

Based on the sample results, Pyrrhobryum produced a larger mean zone of inhibition than Ectropothecium. This may indicate that Pyrrhobryum has stronger antibacterial activity against Escherichia coli.

(Note: In the example presented, the positive control was not included in the main interpretation because its purpose was only to confirm that the antibacterial assay was functioning properly. In antibacterial investigations, positive control has a known antibacterial agent that is expected to inhibit bacterial growth. It serves as a standard reference rather than the focus of comparison. For example, if the positive control produces a clear zone of inhibition, it confirms that the experimental setup can detect antibacterial activity. Likewise, if the negative control shows no inhibition, it confirms that the observed effects are caused by the moss extracts themselves. Therefore, the interpretation focused mainly on comparing Pyrrhobryum and Ectropothecium because these were the actual experimental treatments being investigated.

Step 6 – Accept or Reject the Hypothesis

In the final step, the researcher uses the analyzed results and statistical test results to determine whether the gathered evidence supports or rejects the hypothesis. Statistical tests help determine if the observed differences between treatments are significant and not due to chance alone. This allows the researcher to make evidence-based scientific conclusions.

It is important to remember that the mean alone cannot determine whether a hypothesis should be accepted or rejected. The mean only shows the average result of each treatment. To make a scientific conclusion, researchers use statistical tests to check if the observed differences are truly significant. Examples of statistical tests include the t-test, which is commonly used to compare the means of two groups, and Analysis of Variance (ANOVA), which is used when comparing three or more groups.

Example:
The computed mean showed that Pyrrhobryum produced a larger zone of inhibition than Ectropothecium. After performing a t-test, the researcher found that the difference between the two moss genera was significant. Therefore, the researcher rejected the null hypothesis and accepted the alternative hypothesis.
This means that Pyrrhobryum and Ectropothecium have significantly different levels of antibacterial activity against Escherichia coli.

Remember, young researchers, successful investigations do not only depend on conducting experiments properly. They also depend on how carefully researchers organize, analyze, and interpret gathered data using statistics and hypothesis testing. Through these processes, scientists can determine whether results are truly significant, making scientific investigations more reliable, correct, and evidence based.

PLATE 54. MAKE RESEARCH MISSION CLICKABLE. ADD CLCIKABLE BUTTON TO START THE ACTIVITY



Welcome, young researchers!
You are now officially part of the **Moss Detectives Research Team**. A mysterious bacterial outbreak has affected a nearby community laboratory, and scientists need your help to identify which moss genus may contain stronger antibacterial properties.
Your mission is to investigate the antibacterial activity of the moss genera *Pyrrhobryum* and *Ectropothecium* against *Escherichia coli* using scientific reasoning, statistics, and hypothesis testing.

Will your team successfully identify the stronger antibacterial moss?
Let the investigation begin!

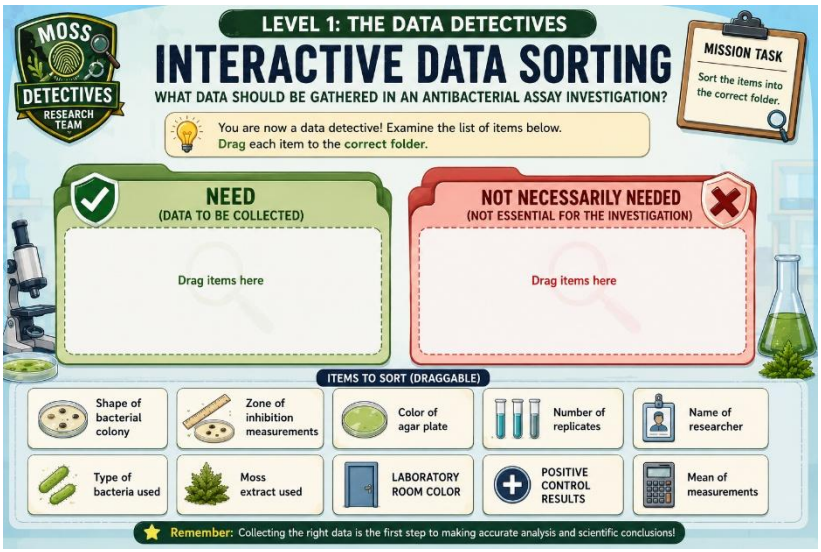
PLATE 55.
Drag and drop activity. Dawaton ra sa boxes bisan wrong ila answer peru automatic mucheck ug muhatag sa ila scores base sa number of correct items. After mucheck magpakita sa correct the arrangement. Proceed na dayun sa next activity



Correct Answer:

1. **Identify the Data to be Gathered**
2. **Formulate the Hypothesis**
3. **Organize the Gathered Data**
4. **Choose the Appropriate Statistical Tool**
5. **Compare and Intepret the Results**
6. **Accept or Reject the Hypothesis**

PLATE 56.



IMPORTANT DATA
Zone of inhibition measurements
Number of replicates
Type of bacteria used
Moss extract used
Positive control results
Mean of measurements

NOT IMPORTANT DATA
- Shape of bacterial colony
- Color of agar plate
- Name of researcher
- Laboratory room color

PLATE 57. ADD NEXT BUTTON PERU DILI MAKANEXT IF WALA MAGSUBMIT



Null Hypothesis (H_0)
There is no significant difference in the antibacterial activity of the 100% and 50% concentrations of *Pyrrhobryum* moss extract against *Escherichia coli*.

Alternative Hypothesis (H_1)
There is a significant difference in the antibacterial activity of the 100% and 50% concentrations of *Pyrrhobryum* moss extract against *Escherichia coli*.

PLATE 58.

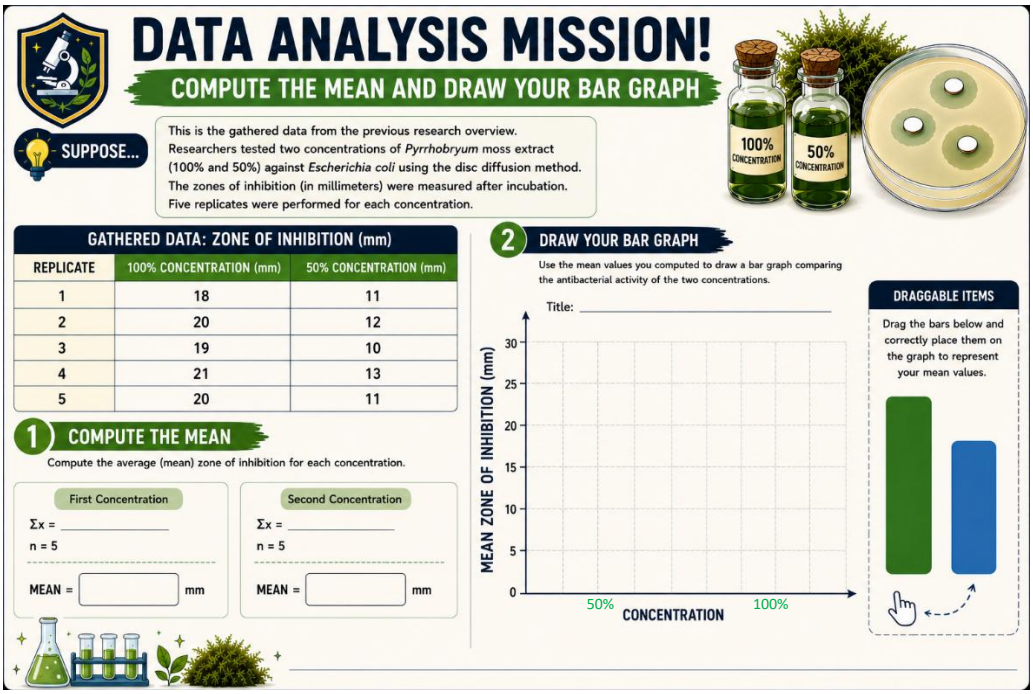
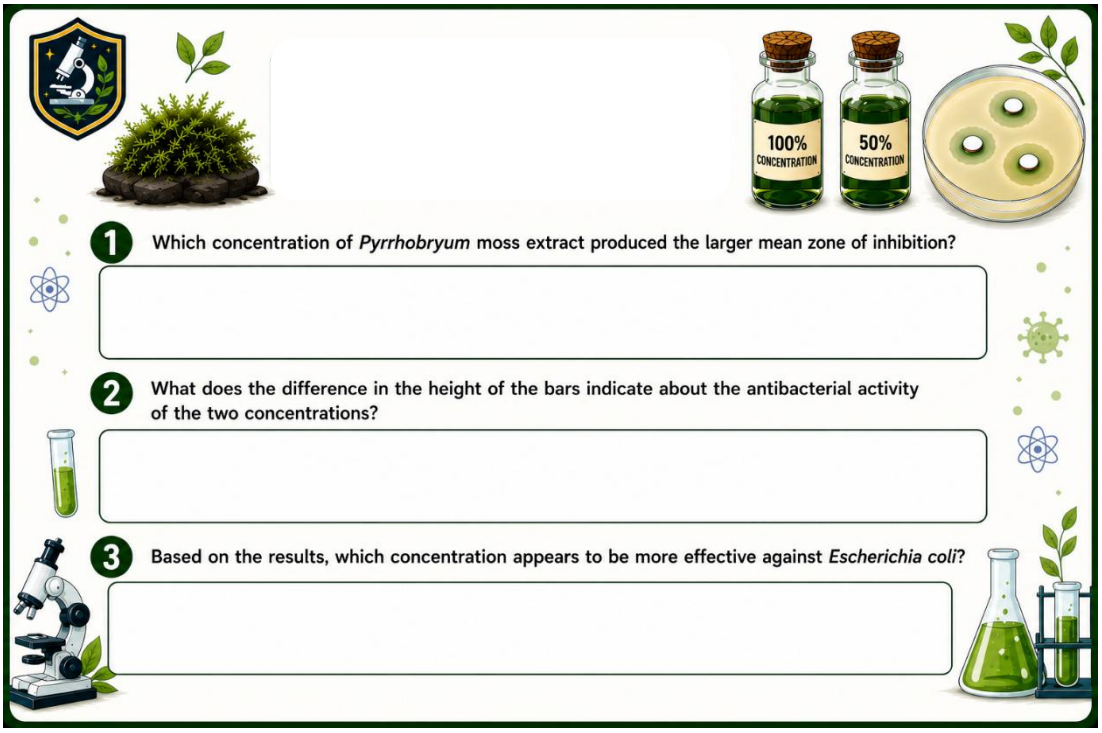


PLATE 59:



ANSWERS:

1. **100% concentration (2 points as long as ma mention ang 100%)**
2. The difference in the height of the bars indicates that the two concentrations have different levels of antibacterial activity. A taller bar means stronger antibacterial activity and a larger zone of inhibition. (5 points for complete answers – same thought)
3. **Based on the results, the 100% concentration appears to be more effective against *Escherichia coli*. (2 points as long as ma mention ang 100%)**

PLATE 60: ANG BATA MAGCLIK LANG WHETER REJECT OR NULL. IF E CLICK ANG WORNG NA ANSWER MU PULA TAS MUMARK UG X, IF MUCLICK UG CORRECT ANSWER, MU GREEN AND MU MARK UG CHECK. THEN MAGWRITE DAYUN NAG BATA SA ILANG CONCLUSION.
ADD CLICKABLE BUTTON FOR MY TAKEAWAY QUESTION



ANSWER: **REJECT**
CONCLUSION: **The results of the investigation showed that the 100% concentration of *Pyrrhobryum* moss extract produced a larger mean zone of inhibition compared to the 50% concentration against *Escherichia coli*. This indicates that the higher concentration exhibited stronger antibacterial activity. Therefore, the researchers reject the Null Hypothesis (H_0) because there is a significant difference between the antibacterial activities of the two concentrations.**

CONDITION FOR SCORING ESSAY
5 Points – Excellent Conclusion
The learner:

- correctly states that the **100% concentration** showed stronger antibacterial activity;
- mentions that there is a **significant difference** between the two concentrations;
- correctly states that the **Null Hypothesis (H_0) is rejected**;
- explains that a larger zone of inhibition indicates stronger antibacterial activity;
- writes a clear and scientifically correct conclusion.

4 Points – Very Good Conclusion
The learner:

- correctly identifies the more effective concentration;
- mentions either the significant difference OR rejection of the null hypothesis;
- provides a mostly accurate scientific conclusion with only minor missing details.

3 Points – Good Conclusion
The learner:

- identifies the correct concentration;
- shows partial understanding of the results;
- may not clearly mention statistical significance or the hypothesis decision;
- conclusion is understandable but incomplete.

2 Points – Fair Conclusion
The learner:

- shows limited understanding of the investigation;
- may mention antibacterial activity but incorrectly explains the results;
- conclusion lacks clarity and important scientific ideas.

1 Point – Poor Conclusion
The learner:

- gives a very unclear, incorrect, or incomplete conclusion;
- shows little understanding of the results or hypothesis testing.

0 Point – No Valid Response

- No answer given; or
- response is unrelated to the investigation.

PLATE 60:

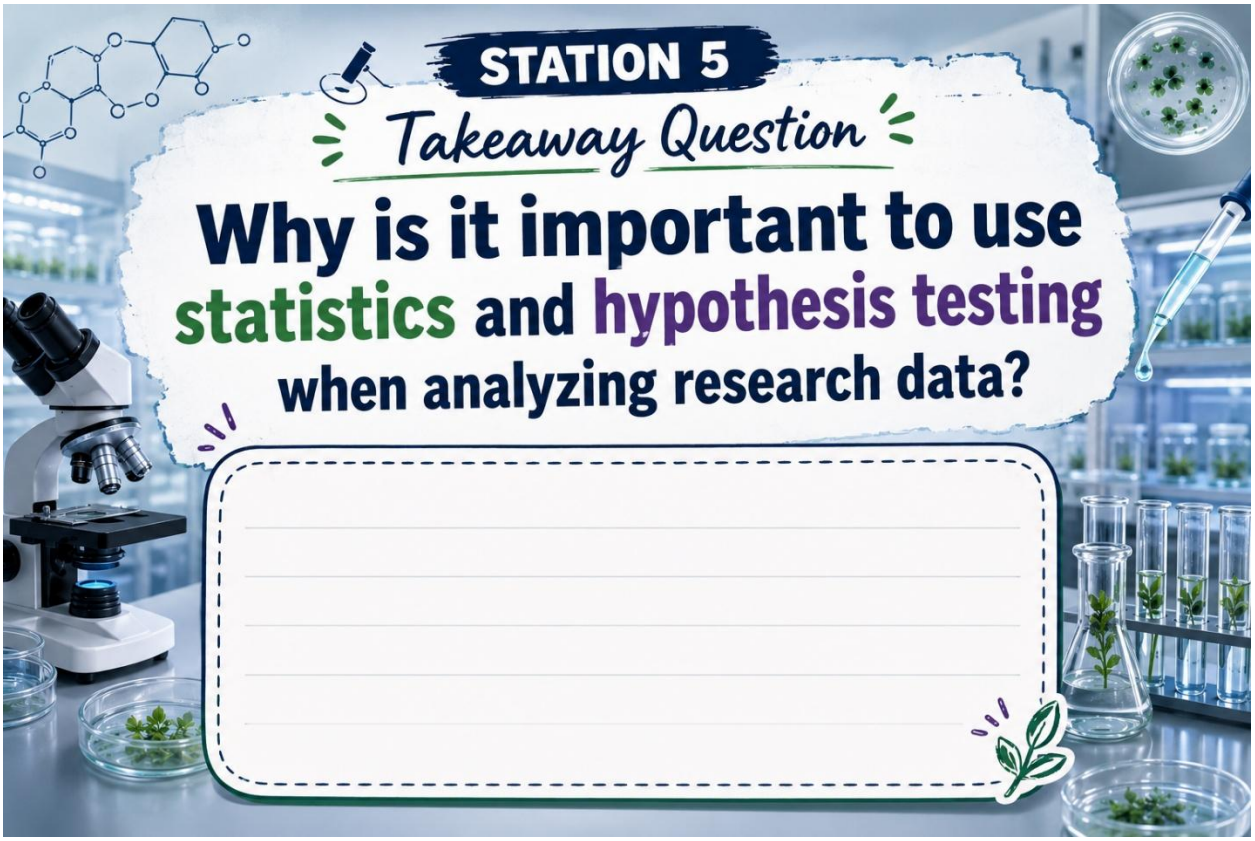
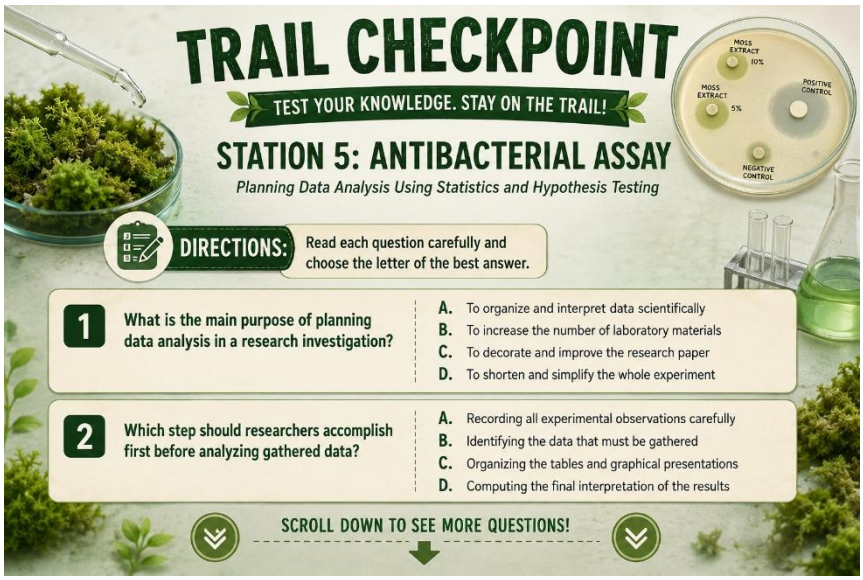


PLATE 61. MAKE THE TRAIL CHECKPOINT CLICKABLE



PLATE 61.



Multiple Choice Test

1. What is the main purpose of planning data analysis in a research investigation?
   A. To organize and interpret data scientifically
   B. To shorten and simplify the whole experiment
   C. To decorate and improve the research paper
   D. To increase the number of laboratory materials

2. Which step should researchers accomplish first before organizing gathered data?
   A. Identifying the data that must be gathered
   B. Recording all experimental observations carefully
   C. Computing the final interpretation of the results
   E. Organizing the tables and graphical presentations

3. Why is identifying the needed data important before conducting the experiment?

A. To prepare the bacterial cultures properly
B. To identify measurements needed in research
C. To determine the treatment groups accurately
D. To organize experimental materials systematically

4. Which statement best describes a scientific hypothesis?
A. A summary of gathered measurements
B. A prediction based on scientific reasoning
C. A conclusion formed after experimentation
D. A detailed explanation of laboratory methods

5. Why do researchers organize gathered data into tables?
A. To compare experimental results more easily
B. To calculate all measurements without statistics
C. To improve the appearance of laboratory reports
D. To prevent recording errors during experimentation

6. Why is the positive control included in the investigation?
A. To compare the treatments carefully
B. To provide additional bacterial samples
C. To confirm the assay functions correctly
D. To increase the inhibition zone measurements

7. What does a larger zone of inhibition usually indicate?
A. Smaller bacterial colonies produced
B. Weaker antibacterial activity observed
C. Lower reliability of experimental results
D. Greater effectiveness against bacteria
For question 8-10, refer to the data table below:

Antibacterial Activity Against *Staphylococcus aureus*

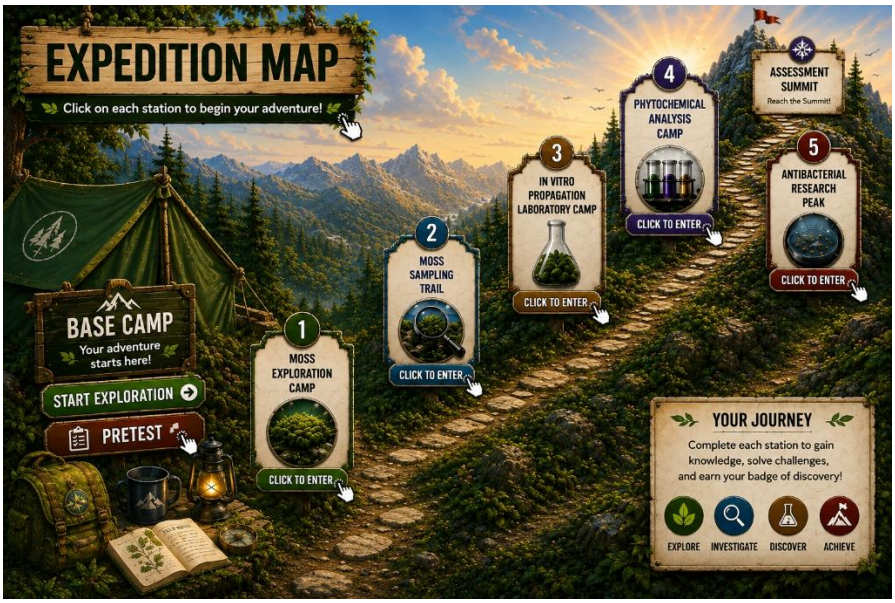
Treatment	Replicate 1	Replicate 2	Replicate 3	Replicate 4	Replicate 5
Pyrrhobryum	16 mm	17 mm	15 mm	16 mm	16 mm
Ectropothecium	10 mm	11 mm	10 mm	12 mm	11 mm

8. **What is the mean zone of inhibition produced by *Pyrrhobryum*?**
A. 14.0 mm B. 15.0 mm C. 16.0 mm D. 17.0 mm

9. Based on the computed means, which moss genus showed stronger antibacterial activity against *Staphylococcus aureus*?
A. Pyrrhobryum showed stronger activity
B. Ectropothecium showed stronger activity
C. Both moss genera showed equal activity
D. Neither moss genus showed antibacterial activity

10. If the researcher performed a t-test and found that the difference between the two moss genera was significant, what should happen to the null hypothesis?
A. Accept the null hypothesis completely
B. Reject the null hypothesis immediately
C. Repeat the experiment without statistics
D. Ignore the hypothesis during interpretation

PLATE. READY TO CLICK NA ANG ASSESSMENT SUMMIT (MAKITA DIRI ANG PROGRESS BAR NA NAHUMAN NA ANG 5 STATIONS



SUMMATIVE TEST

1. A student needs to measure the shoot height of a moss sample accurately. Which instrument should be used?

A. Ruler B. Camera C. Beaker D. Stopwatch

2. Which of the following is an example of a sample?
   A. All mosses on the mountain
   B. All organisms in the ecosystem
   C. Every plant species in the forest
   D. Mosses collected from one small area
3. Which statistical tool is commonly used to determine the average value?
   A. mode B. mean C. median D. ANOVA
4. **What laboratory process is used to identify bioactive compounds present in mosses?**
   A. Soil Profiling
   B. Ecological Mapping
   C. Photosynthesis Analysis
   D. Phytochemical Screening

5. In the antibacterial assay investigation, what will be measured by the researchers?
   A. The amount of moss collected
   B. The temperature inside the room
   C. The diameter of the zone of inhibition
   D. The number of bacteria found in the plate
6. Which study title is an example of descriptive research?
   A. Identifying moss species found on tree trunks
   B. Comparing moss abundance in two forest areas
   C. Testing the effect of nutrient solutions on moss growth
   D. Determining the relationship between moisture and moss diversity

7. **A researcher tests the antibacterial activity of moss extracts against bacteria. Which research design should be used?**
   A. Experimental Research
   B. Descriptive Research
   C. Correlational Research
   D. Quasi-Experimental Research
8. A group collected measurements using different instruments incorrectly. What is the most likely effect on their investigation?
   A. Faster moss growth
   B. More colorful cultures
   C. Less reliable research data
   D. More laboratory equipment
9. **Which research design involves manipulating variables under controlled laboratory conditions?**
   A. Descriptive Research
   B. Correlational Research
   C. Experimental Research
   D. Quasi-Experimental Research
10. Which step should researchers accomplish first before organizing gathered data?
   A. Identifying the data that must be gathered
   B. Recording all experimental observations carefully
   C. Computing the final interpretation of the results
   D.Organizing the tables and graphical presentations

11. Which example below is considered quantitative data in the investigation?
   A. Name of the researcher conducting the study
   B. Type of bacteria used during experimentation
   C. Color appearance of the prepared agar plates
   D. Measurement of the inhibition zone in millimeters
12. Why is choosing the correct instrument important in scientific investigations?
   A. To finish slowly
   B. To avoid observations
   C. To make the laboratory colorful
   D. To collect accurate scientific data
13. Why is organizing collected data in tables important during scientific investigations?
   A. To make the report colorful
   B. To collect the data in a correct way
   C. To make the report longer and better
   D. To compare and analyze data easily
14. Why do researchers usually perform several replicates in an experiment?
   A. To shorten the duration of the investigation
   BTo increase the reliability of gathered results
   C. To reduce the preparation of laboratory tools
   D. To replace the original hypothesis completely

15. What is the main purpose of planning data analysis in a research investigation?
   A. To organize and interpret data scientifically
   B. To shorten and simplify the whole experiment
   C. To decorate and improve the research paper
   D. To increase the number of laboratory materials

16. What does the mean represent in scientific investigations and experiments?
   A. The middle value after arranging all scores
   B. The conclusion written after data interpretation

C. The smallest value among all measurements taken
D. The average obtained from repeated measurements

For numbers 17–20. Refer to the hypothetical data below about the *in vitro* propagation of two moss species.

Researchers observed the number of new shoots produced by two moss species during 4 weeks of *in vitro* propagation.

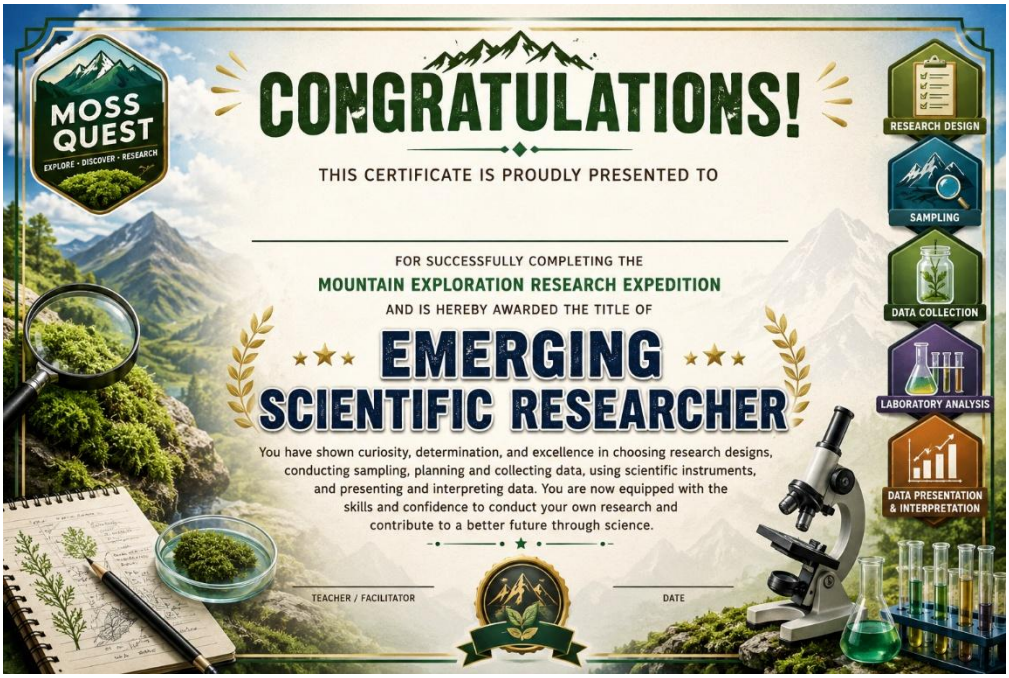
Week	Moss Species A	Moss Species B
Week 1	5 shoots	7 shoots
Week 2	8 shoots	11 shoots
Week 3	11 shoots	15 shoots
Week 4	14 shoots	20 shoots

17. Which statement best explains the trend shown by Moss Species B from Week 1 to Week 4?
A. Its growth changed randomly with no clear pattern
B. Its growth steadily increased over the 4-week period.
C. Its number of shoots gradually decreased each week.
D. . Its growth remained constant throughout the experiment.

18. Which type of graph is most appropriate for presenting the growth of the two moss species over 4 weeks?
A. Pie graph
B. Bar graph
C. Line graph
D. Scatter plot

19. Which statement best compares the growth of Moss Species A and Moss Species B during the 4-week *in vitro* propagation experiment?
A. Both moss species produced the same number of shoots each week.
B. Moss Species B consistently produced more shoots than Moss Species A.
C. Moss Species A consistently produced more shoots than Moss Species B.
D. Moss Species A and B showed decreasing growth throughout the experiment.

20. If the experiment continued for two more weeks and the trend remained the same, what is the most reasonable prediction for Moss Species A?
A. Its number of shoots would suddenly decrease
B. Its growth would likely continue increasing gradually.
C. Its growth would remain exactly at 14 shoots forever.
D. Its growth would become lower than Week 1 results.



ADD ANOTHER BUTTON FOR REFERECES

Asakawa, Y., & Ludwiczuk, A. (2018). Chemical constituents of bryophytes: structures and biological activity. *Journal of Natural Products*, *81*(3), 641–660. https://doi.org/10.1021/acs.jnatprod.6b01046

Bevans, R. (2023, June 22). *Hypothesis testing: A step-by-step guide with easy examples*. Scribbr. https://www.scribbr.com/statistics/hypothesis-testing/

Glime, J.M. (2017). *Meet the bryophytes.* Michigan Technological University; International Association of Bryologists. https://digitalcommons.mtu.edu/bryophyte-ecology1/

Maheshwaran, L., Nadarajah, L., Senadeera, S. P. N. N., Ranaweera, C. B., Chandana, A. K., & Pathirana, R. N. (2024). Phytochemical testing methodologies and principles for preliminary screening/qualitative testing. *Asian Plant Research Journal, 12*(5), 11–38

SurveyMonkey. (n.d.). *Types of quantitative research: Examples + tips*. SurveyMonkey. https://www.surveymonkey.com/market-research/resources/quantitative-research-types